



Research
AI for Smart Agriculture—Review

Extended Reality as a New Farming Tool for Enhancing Smart Agriculture and Precision Farming

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ABSTRACT

Extended reality (XR) is increasingly being explored as a new farming tool that links digital information with physical agricultural environments. Although studies have reported the potential of XR-based human–computer interfaces for agricultural decision-making, operations, and training, the technological pathways, effectiveness, and interdisciplinary implications of these approaches remain incompletely synthesized. This scoping review examines XR application models, enabling technologies, and implementation across crop monitoring, livestock and aquaculture management, agricultural machinery, and education. The reviewed studies indicate that XR can support task-specific perception–analysis–visualization–guidance workflows and may facilitate a shift from experience-based to data-informed decisions, with improvements in precision, efficiency, and safety reported under the tested conditions. We further map the XR technology spectrum and its integration with key enabling technologies, highlighting how these factors jointly shape system performance and field-deployment potential. Major barriers include outdoor adaptability, deployment cost, software integration, data privacy, and limited long-term validation in real agricultural environments. Further progress requires technological integration, interdisciplinary collaboration, cost control, and user-centered design. This review provides researchers, developers, and practitioners with a roadmap for XR solutions that address practical agricultural needs and support greener, more intelligent, and sustainable production.

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1. Introduction

The global population is projected to reach 9.2 billion by 2050, with 66% living in urban areas, increasing the pressure on agricultural systems to meet food demand while maintaining food security and reliable supply [1]. However, intensifying climate extremes [2,3], rising greenhouse gas emissions [4,5], escalating freshwater scarcity [6], and the diminishment of arable land area are making it increasingly challenging to achieve these demands. Additionally, the aging agricultural workforce [7,8] and labor shortages [9] are becoming more pronounced worldwide,

particularly in developed countries. Taken together, these challenges fundamentally reveal the structural limitations of conventional agricultural models in terms of resource efficiency, environmental sustainability, and system resilience [10].

Against this backdrop, smart agriculture and precision farming, centered on information technology, are regarded as key solutions for agricultural transformation [11]. Underpinned by Industry 4.0 technologies, smart agriculture seeks to increase productivity and optimize resource allocation through the Internet of Things (IoT), big data analytics, cloud computing (CC), artificial intelligence (AI), and robotics. These technologies are also discussed in Agriculture 4.0, Aquaponics 4.0, and agricultural metaverse systems as enablers of productivity, data integration, and resource optimization [1,12,13] that can lead to the reshaping of modern agri-food systems [12,14]. As agriculture becomes increasingly

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data-intensive, however, a critical “last-mile” human-machine interaction gap separates back-end digital intelligence from front-line action. Sensor data, AI-generated recommendations, and robot-status information must still be translated into timely, context-specific operations by farmers, technicians, or machinery operators [15]. This gap is evident when users must alternate between physical field tasks and separate dashboards for crop diagnosis, machinery navigation, aquaculture monitoring, or training, underscoring the need for interfaces that connect digital information more directly with physical operations. In this review, we use the term “new farming tool” as a conceptual description of extended reality (XR) as such an interface, rather than as evidence that XR is already mature or universally validated. The value of XR technology is therefore assessed through the tasks, prototypes, and empirical outcomes reported in the reviewed studies.

The core value of XR technology in agriculture lies in its ability to create an interactive space that merges real and virtual environments, effectively blurring the boundaries between physical fields and digital scenarios and thereby enhancing and expanding the capabilities of agricultural practitioners [16]. This technology spectrum is commonly categorized into virtual reality (VR), augmented reality (AR), and mixed reality (MR) according to the degree of immersion and virtual-physical integration—a classification widely adopted in recent XR and agricultural XR studies [15–19]. By providing immersive visualization and enabling more intuitive interaction, XR holds the potential to address part of the “last-mile” human-machine interaction challenge. It does so by presenting complex back-end data from AI and IoT systems to front-line workers in a more intuitive form, thereby helping users manage perceptual and information-processing constraints. Among the technologies on the XR spectrum, VR is the most extensively studied in agriculture [16]. It uses computer-generated, fully virtual

environments to provide users with an immersive experience isolated from the physical environment [20,21]. On the other hand, AR uses the real world as a foundation and superimposes digital information onto it, thus enhancing the user’s perception and understanding of reality [13,22]. Research on MR technology is still in its early stages; MR enables the real-time blending of virtual and real objects and establishes an interactive feedback loop connecting the virtual world, the real world, and the user [23]. Recent agricultural XR and metaverse-related studies have highlighted real-time data visualization, immersive interaction, and context awareness as important advantages of XR interfaces [14,15,24]. In agricultural settings, these capabilities may help extend users’ perception and operational awareness during complex or hazardous tasks, although their effectiveness remains task-specific and requires further field validation.

Despite this potential, XR adoption in agriculture has progressed more slowly than in structured sectors such as healthcare [25,26], manufacturing [27], and education [28]. Agricultural applications are complicated by three interrelated factors: the environment, the objects being monitored and managed, and the tasks involved. More specifically, operational environments are highly unstructured, crops vary substantially in morphology and spatial distribution across growth stages, and agricultural tasks are strongly seasonal and context-dependent. Together, these characteristics make the human-machine-object interaction system unusually complex and impose stringent requirements on the robustness, adaptability, and cost-effectiveness of XR solutions. The conceptual role of XR in linking agricultural challenges with smart and precision farming is summarized in Fig. 1.

Research on agricultural XR has increased across production [29], management [30], and education [31], but its technological pathways, effectiveness, and practical implications remain

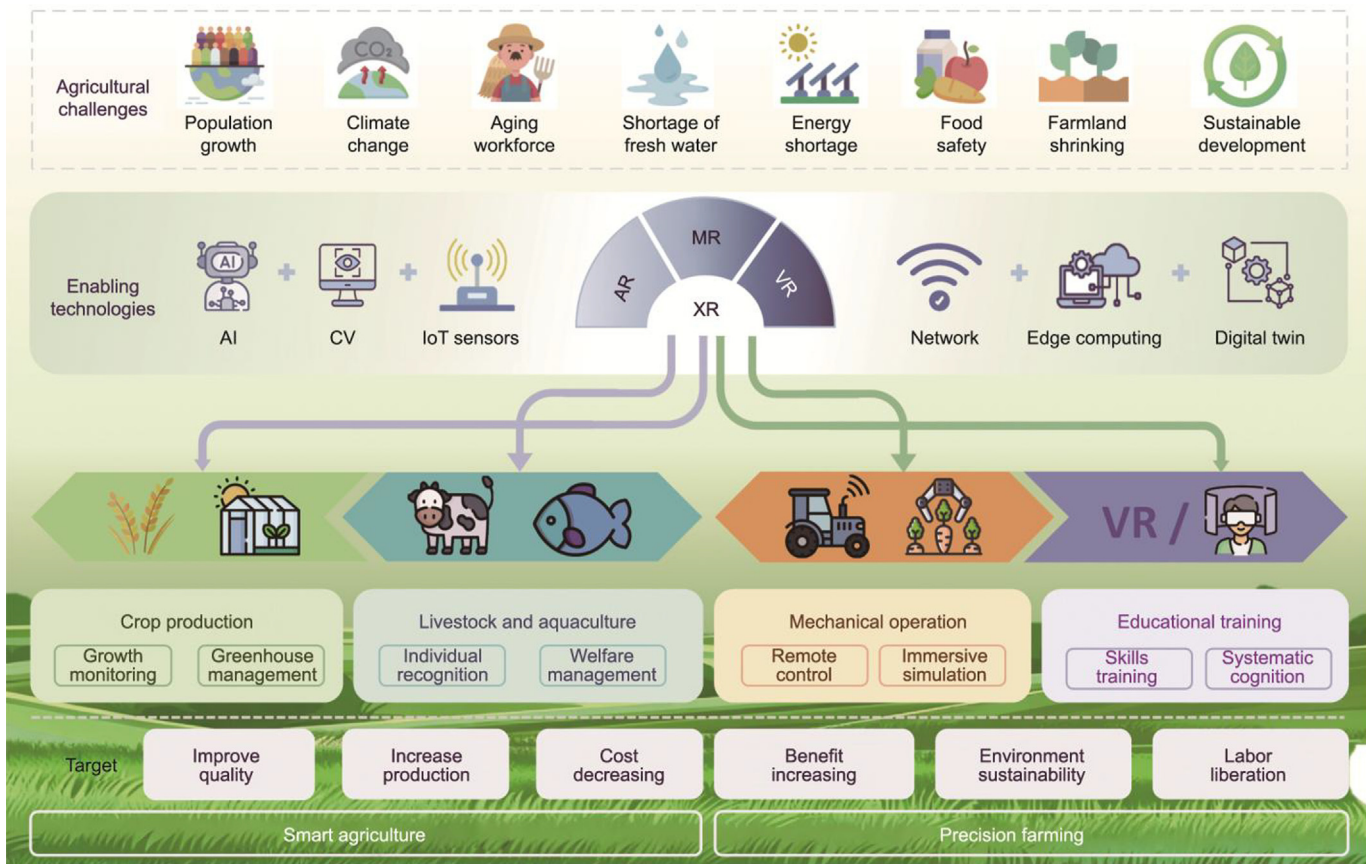


Fig. 1. Integration of XR technologies in smart agriculture: addressing agricultural challenges and enhancing precision farming.

incompletely synthesized. This scoping review therefore examines the XR technology spectrum and enabling technologies, evaluates applications in crop production, livestock and aquaculture management, agricultural machinery, and education and training, and identifies implementation challenges and research priorities for practical adoption and sustainable agricultural development.

To define the scope and literature basis of this review, we searched Web of Science, Scopus, ScienceDirect, SpringerLink, and Google Scholar. The search covered publications available up to 2026, with a focus on studies published from 2018 to 2026, during which period agricultural applications of XR technologies have developed rapidly. Earlier representative studies were also included when they provided important conceptual foundations, technical definitions, or early agricultural XR applications. XR-related terms (“extended reality,” “XR,” “augmented reality,” “AR,” “virtual reality,” “VR,” “mixed reality,” “MR,” and “pure mixed reality”) were combined with agriculture-related terms (“agriculture,” “smart agriculture,” “precision agriculture,” “precision farming,” “smart farming,” “agricultural robot,” “agricultural machinery,” “crop,” “greenhouse,” “livestock,” and “aquaculture”). Boolean queries were also used (e.g., “extended reality” OR “augmented reality” OR “virtual reality” OR “mixed reality”) AND (“precision agriculture” OR “smart farming” OR “agricultural robot”). Because directly relevant studies remain limited, Google Scholar and citation tracking were also used for supplementary searches.

The retrieved literature was screened according to predefined inclusion and exclusion criteria. Studies were included if they focused on XR-related technologies and reported agricultural applications, prototypes, frameworks, technical implementations, or experimental evaluations. Studies were excluded if they were unrelated to agriculture, did not involve XR-based visualization or interaction, focused only on enabling technologies without an XR component, or lacked sufficient technical detail. The included studies were then classified according to XR type, application domain, display device, target object, enabling technologies, and validation context. In addition, VOSviewer was used for keyword co-occurrence analysis to identify research clusters and emerging trends in agricultural XR studies.

2. XR technologies

To classify XR technologies effectively, it is necessary to first introduce the reality–virtuality continuum concept proposed by Milgram and Kishino [32] and improved by Flavian et al. [33]. As shown in Fig. 2, the reality–virtuality continuum frames digital experiences along a spectrum from the purely physical environment to a fully computer-generated one [33]. This continuum is not merely a taxonomic tool; it provides a critical framework for analyzing the hierarchical levels of information delivery that XR technologies can offer in agriculture. These range from unassisted field observation to complete data substitution via VR, with intermediate stages—including AR and MR—blending digital content with the physical world to varying degrees. The core technological pursuit of XR is to achieve seamless technical coverage of this real–virtual fusion zone, enabling natural human interaction with a digitally-augmented space and thereby expanding the boundaries of human perception and operational capability in agricultural settings.

Although definitions of XR subtypes have evolved, current research broadly distinguishes VR, AR, and MR [15,16,34]. For consistency, studies using the term pure mixed reality (PMR) have been grouped under MR when they implemented spatially registered interaction between virtual and physical elements. Finer distinctions within the reality–virtuality continuum were beyond the scope of this review.

These three categories of XR differ in their degrees of integration between real and virtual environments, enabling digital content to extend or replace users’ physical surroundings. In agriculture, VR is mainly suited to simulation, training, digital twins, and remote operation; AR to *in situ* visualization, diagnosis, navigation, and operational guidance; and MR to deeper spatial anchoring and interactive virtual–physical collaboration, although agricultural MR remains less mature than VR and AR.

With its ability to blur the boundary between real and virtual environments, XR has been adopted in retail [35], gaming [36], education [37], catering [38], and healthcare [39]. Agricultural adoption initially lagged because of unstructured environments, limited specialized hardware, high deployment costs, and inadequate perception algorithms. More recently, consumer-grade VR and AR devices have lowered entry barriers and created opportunities in precision farming and smart agriculture. Reported agricultural applications include crop cultivation [40], livestock farming [41], aquaculture [16], remote machinery operation [42], education and training [43], and farmland planning [44,45]. XR-based educational and visualization tools may also attract young people to agriculture [43], potentially supporting the sector’s long-term sustainability.

Fig. 3 summarizes country-level publication output, normalized citation performance, publication recency, and observed coauthorship patterns in agricultural XR research. The United States and China had the largest document outputs, followed by India, Italy, the United Kingdom, and Australia. However, countries such as the Netherlands, Greece, and Austria showed relatively high average normalized citation scores despite their smaller publication volumes. Average normalized citations were calculated by dividing the citation count of each publication by the mean citation count of all publications in the retrieved corpus published in the same year and then averaging these normalized values for each country; thus, a value of 1.0 represents the same-year average within the retrieved corpus. Bubble area represents total link strength—that is, the sum of a country’s observed pairwise coauthorship link strengths with all other country nodes retained in the analyzed network—while bubble color indicates average publication year. Fig. 3(b) shows the upper triangle of the country-pair matrix, with darker cells indicating stronger observed coauthorship. These indicators characterize the retrieved corpus and should not be interpreted as a complete record of international collaboration or as direct evidence of technological or field-deployment maturity.

2.1. Virtual reality

Within the reality–virtuality continuum, VR lies near the “virtual environment” endpoint and creates interactive spaces through computer-generated three-dimensional (3D) models. It is among the most mature and extensively studied XR technologies [19], with conceptual origins often traced to Sutherland’s 1965 work [46]. Closed displays and multimodal interfaces make the virtual environment the primary object of interaction, producing a sense that virtual objects are present and giving VR distinct advantages when users must be separated from the physical environment [47].

The bibliometric patterns in Fig. 2 indicate growing interest in agricultural VR across automation, precision agriculture, education, and human–computer interaction. Reported topics include machinery simulation, technology training, curriculum-based education, immersive learning, and knowledge transfer [19,48–50]. Recent work has also combined XR interfaces with deep learning for tomato yield and grade prediction, maize disease identification, and cassava disease detection and fertilizer recommendation [51–53]. These studies illustrate the breadth of current exploration but do not by themselves establish field-scale effectiveness.

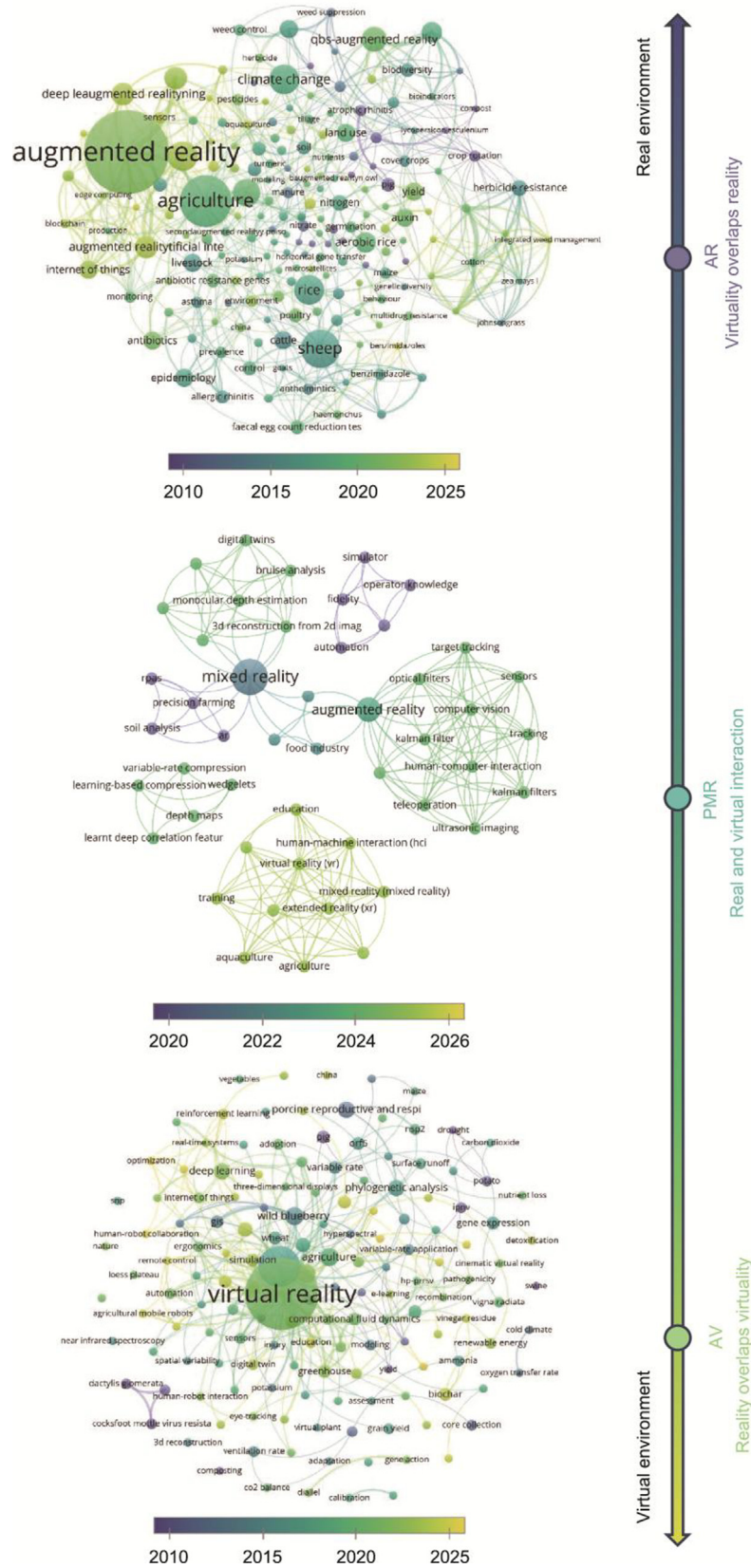


Fig. 2. The reality-virtuality continuum and a keyword co-occurrence analysis of XR technologies.

VR experience quality depends on the hardware ecosystem, the fidelity of the virtual environment, and other factors. Head-mounted displays (HMDs) are among the most common XR

devices [19] and typically present slightly different images to each eye; the brain fuses them to create 3D spatial perception through binocular parallax. Their other core function is head tracking,

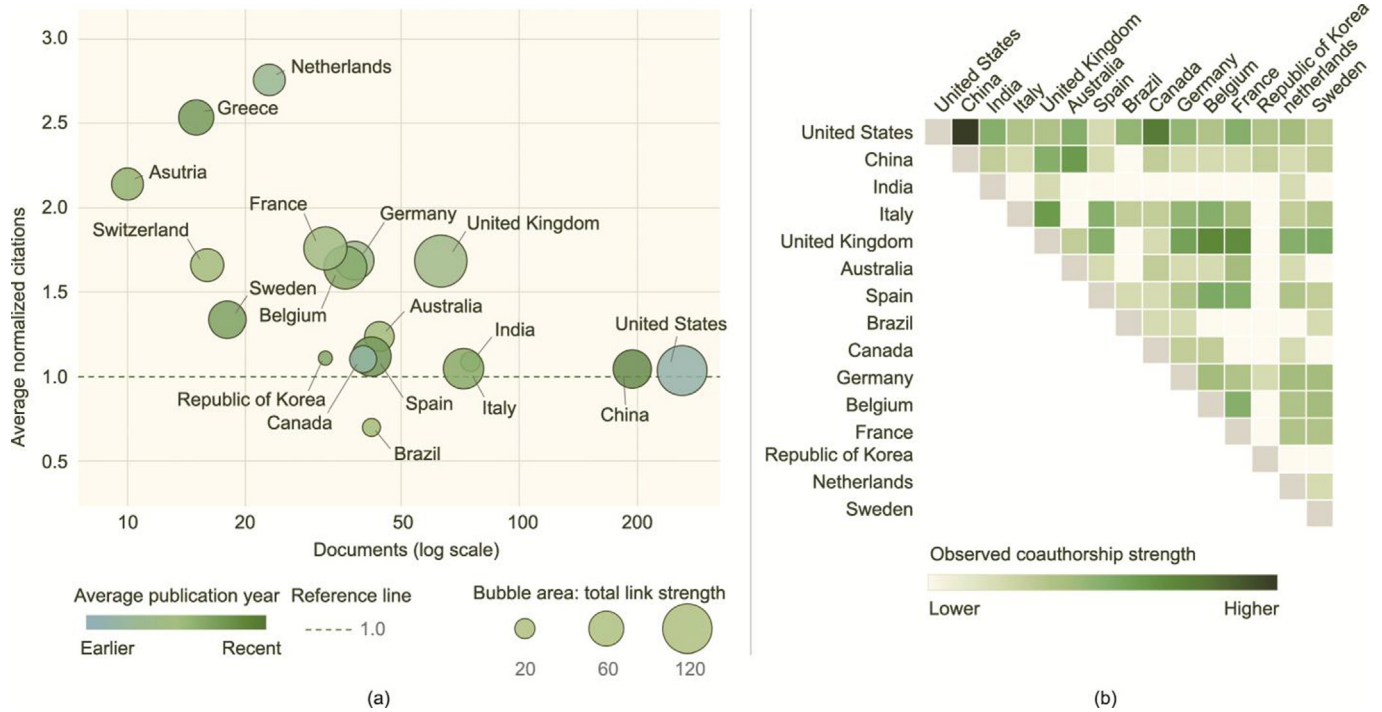


Fig. 3. Country-level bibliometric and observed coauthorship patterns in agricultural XR research. (a) Publication output, average normalized citations, average publication year, and total link strength (bubble area) for selected leading and analytically notable countries. (b) Upper-triangular matrix of observed pairwise coauthorship strength among the 15 countries with the highest document counts.

implemented mainly through inside-out or outside-in approaches [54]. Inside-out tracking combines an HMD's inertial measurement unit (IMU), red-green-blue and depth (RGB-D) cameras, and simultaneous localization and mapping (SLAM) to estimate pose in real time. It allows flexible deployment but requires multisensor fusion to limit sensor errors [55]. Outside-in tracking instead uses external base stations or cameras to locate the HMD [56].

Beyond HMDs, VR systems use peripheral devices to capture user actions and enhance immersion. Handheld controllers commonly provide motion tracking and user input, while HMD-integrated cameras with computer vision (CV) algorithms enable vision-based gesture recognition [19]. Gamepads, motion-capture sensors, and data gloves provide additional input channels by capturing user actions and translating them into system commands [16,57,58]. Haptic devices simulate forces, vibrations, or motion, extending interaction beyond vision and audio [59,60]. In precision agricultural tasks, such feedback supports force-feedback simulation and improves immersion and operational realism [61].

A principal motivation for adopting VR in agricultural research is its capacity for immersive visualization and situational simulation [19]. This capacity supports several application paradigms. First, VR enables remote operation and supervision of agricultural robotic systems, reducing distance constraints and providing an immersive interface through which one operator may monitor multiple machines [62]. Second, digital-twin systems can provide model- and data-driven views of large-farm conditions and support the control of irrigation and ventilation [30]. Finally, VR can provide risk-free, repeatable, and potentially cost-effective environments for machinery operation, farm-system learning, horticultural and landscape-management training, and classroom education [63–65].

Notwithstanding this potential, several core challenges impede broader VR adoption in agricultural settings. These include the substantial resources required for high-fidelity modeling of complex, dynamic agricultural environments; the ergonomic

discomfort of current HMDs during prolonged use; the dependence of real-time teleoperation and distributed multisensor data exchange on reliable, ultra-low-latency communication networks; and the vulnerability of optical tracking systems under challenging outdoor lighting conditions [20,66].

2.2. Augmented reality

AR lies near the real-environment end of the continuum and overlays digital content onto the physical environment [23,67]. By preserving users' view of their surroundings, it enables direct, *in situ* interaction with contextual data and guidance. In smart and precision farming, AR is increasingly studied for real-time visualization, resource management, environmental monitoring, and operational support [16,57,68]. These functions can place crop-growth, soil-quality, or management information at the point of action, although their effectiveness depends on reliable registration and outdoor display performance.

AR display systems are commonly categorized as monitor-based, optical see-through, or video see-through [57]. Monitor-based AR overlays digital information on a camera feed displayed on an external screen; it is simple and inexpensive but offers limited immersion [69]. Optical see-through AR superimposes virtual images via transparent optical combiners or waveguides while preserving a direct view of the physical world, although strong outdoor illumination can reduce virtual-content visibility [70]. Video see-through AR captures the environment with head-mounted cameras, digitally integrates virtual content, and presents the composite through opaque near-eye displays. It supports algorithmic compensation for changing illumination and richer visual effects but demands greater camera and processing capability and may introduce latency or reduce perceptual realism.

A robust agricultural AR experience depends on precise and stable registration—the process of anchoring virtual content accurately to the physical world. This relies on cameras, IMU, the global

positioning system (GPS) [71], and visual SLAM algorithms [72], often used in hybrid configurations [69]. The task requires not only determining the user's global position but also continuously recognizing and tracking specific objects within the environment. In agricultural settings, GPS signals can be unreliable, and visual tracking algorithms must contend with dynamic, non-rigid objects such as wind-swayed plants and evolving canopy structures, making stable registration difficult.

In smart agriculture, AR can support field inspection, data visualization, and operational guidance. It can overlay pest and disease warnings, soil-moisture data, and fertilization prescriptions on corresponding crops or plots, or display health and breeding information on individual animals, thereby supporting more precise management and intuitive decision-making [57]. Agricultural AR nevertheless presents several challenges: Variable outdoor illumination can interfere with visual recognition; unstructured farmland imposes high demands on object identification and tracking; and wind-induced plant movement, dust, mud, and rain can affect device stability and durability.

2.3. Mixed reality

MR seeks to establish a real-time, bidirectional interaction loop among the user, virtual objects, and the physical environment [73,74]. Unlike AR systems, which primarily overlay information, MR requires seamless virtual-physical blending, real-time operation, and physically plausible interaction [23]. Its central challenge is environmental understanding: Virtual objects must respond coherently to physical objects and user actions, which requires robust sensing and real-time simulation [73].

MR relies on CV, tracking, and sensors to support environmental understanding, object recognition, and real-time positioning. MR systems may use head-mounted, handheld, monitor-, or projection-based displays [75]. Head-mounted and related see-through implementations can follow optical or video see-through approaches. Optical see-through systems preserve a direct view of the physical environment through transparent combiners while superimposing digital content [76]. Video see-through systems instead capture the environment with cameras, merge the footage digitally with virtual elements, and present the composite view on a display. These hardware and interaction mechanisms substantially overlap with AR, and AR devices are often used in MR-oriented studies.

In this review, AR and MR are distinguished according to the depth of virtual-physical integration rather than the display device used. Systems are classified as AR when they mainly superimpose digital information, labels, instructions, or data panels onto the real environment without physically consistent or bidirectional interaction between virtual and real objects. They are classified as MR when they involve deeper virtual-physical integration, typically including spatial mapping, environmental understanding, persistent spatial registration, and object-level or real-time interaction among virtual objects, physical objects, and users. Therefore, the use of an AR HMD alone does not automatically define a system as AR or MR; the classification depends on the functional interaction level implemented in the study.

With advances in computation and display hardware, MR applications have expanded beyond entertainment into education, marketing, safety training, and other interactive settings [36,77–79]. In controlled-environment agriculture (CEA), MR could spatially anchor equipment data to physical devices and support gesture-based parameter adjustment. In remote collaboration, specialists could guide onsite maintenance through shared spatial annotations or virtual avatars [80]. Such applications may support more natural human-machine collaboration in agricultural cyber-physical systems.

2.4. Common and key enabling technologies

XR performance and agricultural applicability depend on an integrated stack of enabling technologies. An HMD alone cannot provide a robust XR experience; complementary systems must support perception, intelligence, connectivity, and computation [57]. This section examines CV, AI, communication networks, and edge computing as the principal technological foundations of agricultural XR. The relationships among these enabling technologies and their roles in agricultural XR systems are summarized in Fig. 4.

2.4.1. CV and XR

CV provides the environmental and user understanding required for immersive XR [81,82]. Visual SLAM estimates device poses in real time from onboard cameras [83]; CV-based analysis of HMD camera images enables hand tracking and bare-hand gesture interaction [84]; and vision-based eye tracking supports foveated rendering and gaze-based input [85]. Together with computer graphics, CV supports lighting estimation, shadow computation, texture mapping, and geometric processing for rendering. Unstructured and dynamic agricultural environments complicate these perception tasks: SLAM must handle repetitive crop rows and changing canopies, while object recognition must remain robust to lighting variation, occlusion, and growth-stage changes. Addressing these domain-specific constraints is essential for stable and realistic XR experiences.

2.4.2. AI and XR

Advances in AI are expanding data-analysis capabilities in smart agriculture. Integrating AI with XR can support smart and precision agricultural systems by linking analytical outputs with contextual interfaces. AI provides the analytical foundation for processing large-scale, complex, and multimodal agricultural data, including animal-behavior monitoring, livestock localization, crop-stress visualization, and canopy-instance segmentation [86–89]. XR serves as an immersive interaction interface that presents AI-generated results to users in an intuitive manner. Machine-learning and deep learning algorithms can integrate and analyze such data for diagnosing pests, diseases, and weeds [90–94]; crop counting and grading [53,95,96]; and yield prediction [97–99]. The combination of AI and XR may therefore reduce subjectivity in experience-dependent approaches and improve the responsiveness and flexibility of conventional decision-support systems.

Recent advances in large language models may further extend AI–XR integration by interpreting natural-language queries, integrating agricultural knowledge, and supporting extension services or decision guidance [100–102]. XR interfaces could translate such recommendations into contextual visual guidance, potentially lowering technical barriers to complex agronomic operations. Related studies have also explored integrating XR with AI, vocational training systems, and rural or agricultural digital services to support workflow guidance, knowledge transfer, and operational efficiency in specific application contexts [15,103,104].

2.4.3. Communication networks and XR

Low-latency, high-bandwidth, and reliable connectivity is critical to agricultural XR performance and user experience [105]. Delays in sensory feedback or control commands can disrupt presence and interaction and may cause operational errors or safety risks during remote machinery operation. The scale and topographic variability of farmland make wired deployment costly, increasing demand for wireless, lightweight, and low-power communication systems. As XR moves toward cloud rendering and remote operation, stable and efficient network support becomes increasingly important.

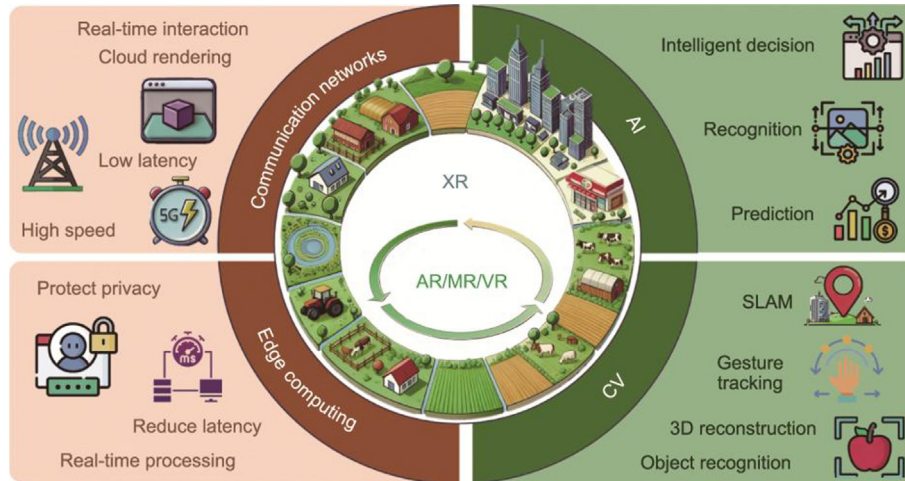


Fig. 4. Key enabling technologies for XR systems in agricultural applications.

In agricultural settings, IoT infrastructure incorporates low-power, wide-area network protocols such as ZigBee, LoRa, and Sigfox, as well as fourth-generation long-term evolution (4G LTE) mobile communication [1]. Fifth-generation (5G) and prospective sixth-generation (6G) mobile networks, which offer millisecond-level latency, gigabit-per-second data rates, and high connection density, may support high-quality XR [106]. Such networks can provide infrastructure for high-definition video transmission, cloud/edge rendering, remote control, and immersive collaboration. However, the reviewed agricultural literature generally treats connectivity as a background condition rather than an explicitly evaluated variable. Few studies isolate the contribution of latency, bandwidth, mobility, or field coverage to XR performance, leaving this an important direction for deployment research.

2.4.4. Edge computing and XR

Complex agricultural XR applications, including whole-farm digital twins, may exceed the rendering and computational capacity of standalone devices [107]. CC can supply additional resources but may introduce latency through network conditions and transmission distance. Edge computing addresses both constraints by moving computation closer to the user [108]. In agricultural XR, it can support high-frame-rate rendering, real-time CV inference and visual overlays, and the physics calculations required for MR interactions. Combined with 5G and lightweight AI models, this edge-AI-network architecture can provide the computational and communication backbone for demanding agricultural XR applications.

In summary, agricultural XR depends on the coordinated operation of CV, AI, communication networks, edge computing, and specialized hardware. This enabling stack supports production monitoring and decision systems [109], remote agricultural machinery operation [42,110], and agricultural education and training [34,43,111]. Its practical value therefore depends on system-level integration rather than any single technology, particularly under heterogeneous field conditions.

3. Applications of XR technologies in agriculture

Across smart agriculture, XR technologies have been explored for diverse operational needs, offering intuitive interaction and visualization. Major application areas include growth monitoring and management, livestock and aquaculture management, agricultural machinery, and education and training. To support cross-domain comparison, this section summarizes the reviewed studies

using a common structure: application category and scenario, target/object, display device, XR type, and reference.

3.1. XR applications in growth monitoring and management

Crop growth monitoring and management are a critical link in agricultural production, directly affecting both yield and quality, and are therefore among the primary concerns for farmers [112]. While CV and AI have revolutionized tasks from harvesting to disease prediction, their impact has been constrained by a fundamental interface gap. Farmers must often alternate between physical inspection and digital screens, disrupting workflow and intuition. XR technology can help bridge this limitation by fusing the digital and physical realms [113–115]. Through augmented and immersive interfaces, XR embeds information processing and decision support directly into the farmer's field of view, establishing a more continuous pipeline from data acquisition to actionable guidance. This section explores how this capability is forging new paradigms in crop monitoring, pest management, and farm system control (Table 1 [34,40,51–53,68,116–145], Fig. 5 [118,120,121,132]).

3.1.1. Crop growth monitoring

Traditional agricultural production remains experience-dependent, leaving field-management efficiency and quality vulnerable to individual and environmental variability. Agricultural AI can reduce this dependence, but its practical value is limited when analytical outputs are separated from field observation. XR addresses this interaction bottleneck by presenting these outputs within the field context, enabling a continuous transition from analysis to crop-management action.

A growth-monitoring system based on XR primarily follows a perception–analysis–visualization–guidance technological approach. This process begins with the collection of crop data through front-end sensors, followed by state analysis using back-end AI models or computer information-processing systems. The analysis results are then overlaid onto the real-world scene in real time through XR devices, in the form of virtual labels, instructions, or other indicators, providing users with intuitive operational guidance.

For instance, in harvesting, Tamura et al. [122] developed an AR HMD-based system for strawberry picking. The system uses a deep neural network (DNN) to identify strawberry maturity and overlay virtual labels on the actual fruit in real time, thereby reducing misjudgments caused by inexperience. Buayai et al. [121] created an AR-assisted system for grape berry thinning, a time-sensitive and

Table 1

A comprehensive summary of applications in growth monitoring and management.

Application category	Application scenario	Target/object	Display device	XR type	Reference
Crop growth and harvesting	Growth monitoring	Plant	Portable device	AR	[116]
	Growth monitoring	Plant	Portable device	AR	[117]
	Crop stem counting	Wheat	AR HMD	AR	[118]
	Crop weight measurement	Tomato	AR HMD	AR	[119]
	Ripeness detection	Strawberry	AR HMD	AR	[120]
	Yield prediction	Tomato	Portable device	AR	[53]
	Agronomic operation support	Grape berry	AR HMD	AR	[121]
	Harvesting assistance	Strawberry	AR HMD	AR	[122]
	Yield monitoring	Strawberry	VR HMD	VR	[123]
	Harvesting support	Tomato	AR HMD	MR	[124]
	Plant breeding and trait evaluation	Canola	VR HMD	VR	[125]
	Crop growth monitoring and simulation	Crop	VR HMD	VR	[126]
	3D digital twin generation of agricultural products	Agricultural products	MR HMD	MR	[127]
Pest and disease management	Crop monitoring	Crop	Portable device	AR	[128]
	Crop disease identification	Plant disease	AR HMD	AR	[129]
	Crop disease identification	Corn	AR HMD	AR	[51]
	Crop disease identification	Plant disease	Portable device	AR	[130]
	Disease detection and fertilizer recommendation	Cassava leaf	Portable device	AR	[52]
	Pest detection and counting	Rice planthoppers	AR HMD	AR	[131]
Field and greenhouse management	Pest detection and counting	Leafminer	AR HMD	AR	[132]
	Intelligent irrigation and control	Irrigation	Portable device	AR	[68]
	Automated irrigation	Greenhouse	Portable device	AR	[133]
	Real-time greenhouse control	Greenhouses	Portable device	AR	[134]
	Greenhouse microclimate management	Greenhouses	Portable device	AR	[135]
	Optimized control and decision-making	Greenhouses	Portable device	AR	[136]
	Greenhouse digital twin	Greenhouses	VR HMD	VR	[137]
	Medicinal crop production management	Cannabis	VR HMD	VR	[40]
	Smart farm monitoring	Greenhouses	Portable device	AR	[138]
	Farm information management	Crop	Portable device	AR	[139]
	Agricultural fieldwork navigation	Fieldwork	VR HMD/portable device	AR	[140]
	Plant species identification	Aromatic herb	Portable device	AR	[141]
	Plant information identification	Plant	Portable device	AR	[142]
	Immersive virtual greenhouse training	Greenhouse	VR HMD	VR	[143]
	Visualization of the greenhouse aerodynamic environment	Greenhouse	VR HMD	VR	[144]
	Immersive viticulture education	Grape	VR HMD	VR	[34]
	Soil sampling	Soil	AR HMD	AR	[145]

In the tables, XR type is assigned according to the functional characteristics of each system rather than the display device name. AR refers to digital information overlaid on the real environment, whereas MR refers to deeper spatially registered and interactive virtual-physical integration.

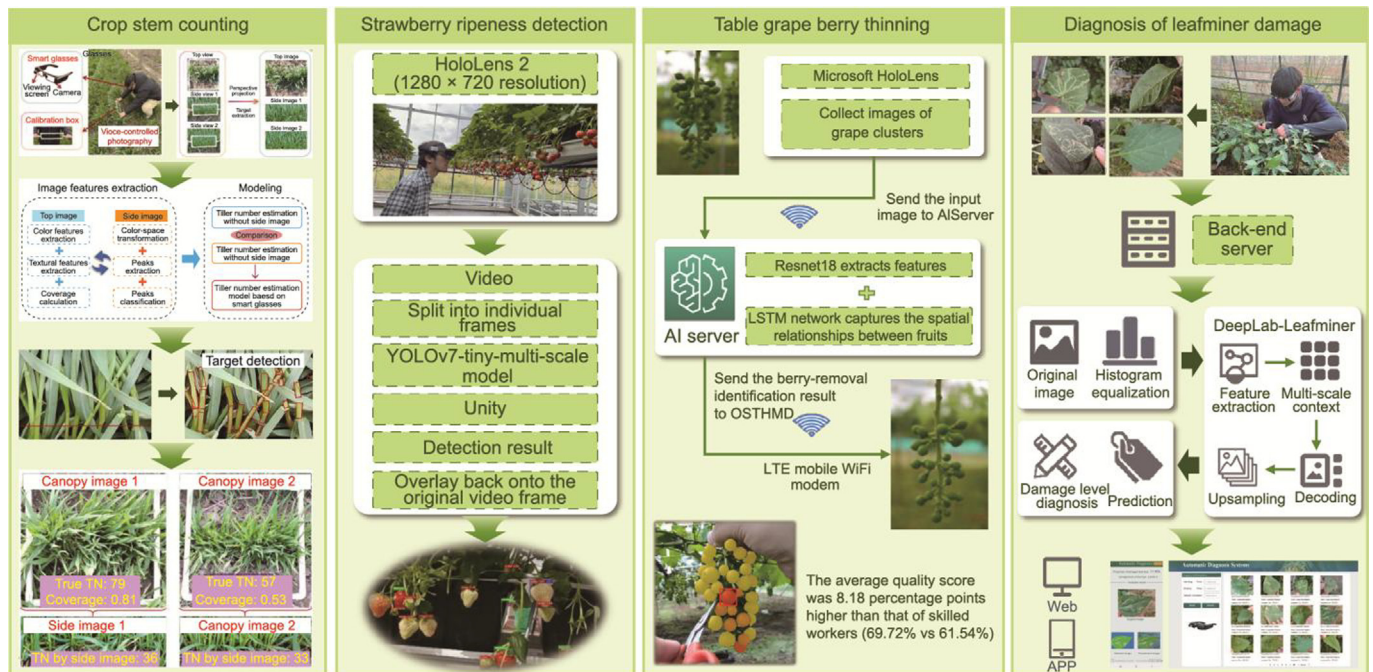


Fig. 5. Overview of agricultural task AR systems: crop stem counting [118], strawberry ripeness detection [120], table grape berry thinning [121], and diagnosis of leafminer damage [132]. TN: tiller number; LSTM: long short-term memory; OSTHMD: Optical See-Through HMDs; APP: application.

technically challenging task. A neural network analyzes images of grape clusters and automatically identifies berries for removal. The AR headset then provides real-time instructions that guide farmers to the selected berries. Experimental results showed that AR-guided untrained farmers achieved higher operational quality than experienced farmers, with an average improvement of 8.18%. For crop yield prediction and grading, Prabhu et al. [53] developed a mobile AR application that uses deep learning to classify and count tomatoes by maturity, providing data for yield forecasting. Users can also grade tomatoes and obtain information on quantity and relevant varieties through the application. Sakha et al. [125] designed a field robot to collect *in situ* data from oilseed rape breeding materials. The data are visualized in a 3D VR environment, facilitating remote observation and the evaluation of numerous varieties by breeding experts.

3.1.2. Pest and disease management

The integration of XR technology with CV and AI offers innovative technical pathways for precise identification and control of crop diseases and pests in the field. Its potential effectiveness lies in its ability to support task-specific workflows that combine real-time diagnostics, visual enhancement, and decision support. XR systems can interact with real-time visual data and provide intuitive visualizations to support users in efficiently completing diagnostic tasks, significantly changing how agricultural technicians perceive and interpret crop health status in the field [146].

In the area of disease detection, Balahadia et al. [51] combined AR devices with a convolutional neural network (CNN) algorithm to develop an in-field solution for the rapid diagnosis of maize diseases. Similarly, Roopa and Bose [130] created a low-cost AR system that uses the You Only Look Once (YOLO) v5 object-detection model to locate leaf regions and a CNN for the semantic segmentation and classification of diseased spots, ultimately overlaying disease information onto the real-world view via AR technology. For cassava leaf disease detection, Prashanth et al. [52] proposed a hybrid deep learning model, Modified Pyramidal Convolutional Shuffle Binary Attention Residual network with Advanced Harris Hawk optimization algorithm (MPCSR-AHH), integrated with AR, enabling real-time detection of four common diseases, such as brown spot and frog-eye leaf spot. This model integrates detection, diagnosis, and fertilizer recommendation into a task-specific decision-support workflow.

In terms of pest monitoring, AR glasses equipped with imaging and display functions have become efficient tools for field surveys. Ye et al. [132] adopted a solution combining AR glasses with a DeepLab-Leafminer model to achieve accurate and real-time assessment of leafminer damage levels. This system effectively addresses technical challenges such as uneven leaf surfaces and difficult segmentation of damaged areas in field conditions, automating the entire process from image acquisition and lesion segmentation to damage-level diagnosis. Additionally, to tackle the inefficiency and high error rates of traditional rice planthopper survey methods, Sheng et al. [131] proposed a combined AR and deep learning approach that uses AR glasses to capture images of insects on whiteboards and employs an algorithm for automatic counting. This method reduced required labor by half while significantly improving survey efficiency and accuracy; it also holds potential for extension to monitoring other pests such as aphids and whiteflies.

3.1.3. Field and greenhouse management

Field and greenhouse management are core agricultural activities, and their complexity increases as operations expand. Crop health and development depend on multiple environmental factors [147]. Although sensor networks can monitor these conditions in real time, their data are typically centralized in control-room sys-

tems. On large commercial farms, the distance between the control rooms and the fields forces managers to move frequently between locations to obtain current information, inspect conditions, and make decisions. This separation impedes management efficiency and makes rapid responses to extreme weather or sudden environmental changes difficult.

To address these challenges, researchers have integrated XR with farm management systems to provide AR visualization and interactive decision support. Katsaros and Keramopoulos [142] developed a mobile AR application that helps farmers identify crops and access relevant knowledge bases, reducing the need to memorize detailed information on pests, diseases, and crop-specific traits. Zheng and Campbell [140] designed an AR system for *in situ* information visualization and field-operation guidance. Without disrupting ongoing production, the system supports decision-making during inspection and management on large farms. Spyrou et al. [40] constructed a VR-based digital twin for medicinal cannabis production. By integrating growth models and environmental parameters, the system enables dynamic visual simulation. The study demonstrated its feasibility in highly regulated, high-standard greenhouse environments. Ulagammai and Moorthy [68] developed a closed-loop sensing-to-control system: An ESP32-based IoT architecture collects soil sensor data, an AR interface visualizes them, and users can interact with the system and operate the irrigation systems.

Beyond direct onsite management, another important advantage of XR is its capacity for simulation, prediction, education, and training. High-fidelity simulations of real environments allow users to evaluate management strategies and operational commands before implementation, helping managers optimize decisions and practitioners develop a deeper understanding of complex agricultural systems. Carruth et al. [143] developed an immersive VR greenhouse simulator for CEA. The system provides a realistic interface for physical operations and CEA system control, offering students hands-on training while avoiding the risk of crop loss through improper management. Kim et al. [144] created an integrated solution combining computational fluid dynamics (CFD) simulation, 3D scanning, and VR interaction. By visualizing airflow through multiple modalities, the system makes otherwise difficult-to-observe greenhouse environmental processes easier to understand. Spyrou et al. [34] proposed a framework integrating VR, a digital twin, and an AI virtual instructor; it can simulate the long-term effects of vineyard management strategies and provide personalized guidance, alleviating the long cycles and high costs of real-world training.

These studies suggest that XR technology not only offers new pathways to address traditional management challenges such as experience dependence and delayed response by improving the efficiency of data interaction and utilization but can also help establish a technical foundation for optimizing resource allocation, training professionals, and advancing the precision production of high-value crops through advanced simulation and training functionalities.

3.1.4. Summary and discussion

Analysis of the three application areas reviewed above yields a unified perception–analysis–visualization–guidance framework for XR-enabled crop growth monitoring and management, linking the physical environment, information processing, and field action (Fig. 6). Sensors integrated into XR devices or deployed in the field form the perception layer, collecting crop and environmental data *in situ*. AI models and other computational methods constitute the processing and decision layer, analyzing these data to assess current conditions, predict changes, and generate decisions or recommendations. The XR interaction layer then converts analytical outputs into intuitive visual cues, overlaying them on the user's

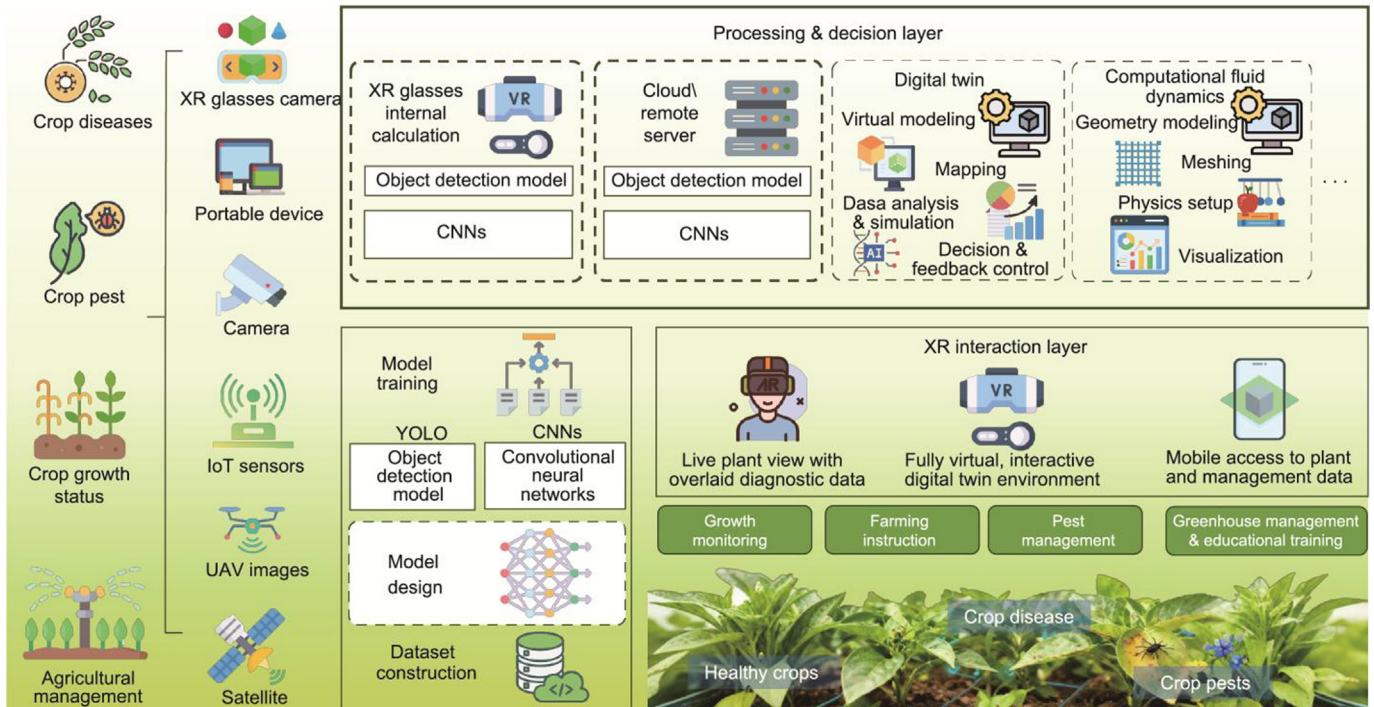


Fig. 6. XR-based agricultural growth monitoring and management system: key components and layers; UAV: unmanned aerial vehicle.

field of view or operational environment in AR/MR systems or presenting them within immersive virtual environments in VR, thereby guiding field action.

This framework illustrates a key potential of XR technology: It can serve as a data interface that helps connect experience-based observation with data-informed decision-making. In this sense, XR compresses the traditionally separated steps of observation, analysis, decision-making, and execution into a unified, intuitive interactive space. Some reviewed studies have reported improvements in specific tasks, such as operation quality or diagnostic efficiency; however, broader claims of cost reduction, quality improvement, and inclusivity remain to be validated at scale. Furthermore, for advanced simulation and training functions, XR technology can extend beyond real-time guidance, offering a promising basis for optimizing resource allocation, training professionals, and advancing the precision production of high-value crops.

Nevertheless, XR applications in crop monitoring and management still present several limitations. Current systems often address task-specific problems, such as maturity detection, variety identification, yield estimation, or disease diagnosis, and few studies have demonstrated multifunctional systems that can be generalized across crops and production scenarios. Some applications also depend on CC for image processing or model inference, which may introduce latency and reduce real-time responsiveness. Although XR-AI systems have achieved promising accuracy in specific crop diagnosis tasks, most have been developed for single crops or controlled environments. Their generalizability to mixed-crop fields, different growth stages, and large-scale open-field environments remains uncertain, particularly under strong sunlight, occlusion, wind-induced plant movement, dust, and variable canopy structures. Therefore, current evidence mainly supports the technical feasibility of XR-enabled crop management, while field robustness and scalability still require further validation.

3.2. XR applications in livestock and aquaculture management

Although the status of animal welfare has been significantly improved in recent years, the livestock industry still faces chal-

lenges such as low data visibility, difficulties in individual animal identification, insufficient welfare-monitoring methods, and low management efficiency [148]. With its capabilities in immersive interaction and virtual-real integration, XR technology offers novel approaches to address these issues, as summarized in Table 2 [149–159].

3.2.1. Livestock farming

In livestock farming, XR applications focus primarily on individual identification, animal health monitoring, and welfare management, supporting the fine-grained management of individual animals (Fig. 7 [150,152,153,155,156]). For identification and data visualization, Zhao et al. [149] integrated GPS and CV in a mobile AR application to track specific dairy cows and overlay their records, addressing inefficient conventional data queries. Pinna et al. [150] combined radio-frequency identification (RFID) with a SmartGlove and an AR HMD. By transmitting RFID data to the HMD via Bluetooth, the system enables rapid animal recognition and real-time visualization of breeding data. Together, these studies show how XR can support precision livestock management. Arkan et al. [152] used deep learning to identify dairy cows and detect estrus, transmitting the results in real time to an AR management platform. In welfare and environmental monitoring, Baltzer et al. [155] developed AR smart glasses that capture thermal-infrared, ultraviolet, and visible red-green-blue (RGB) images to monitor poultry-house conditions in real time.

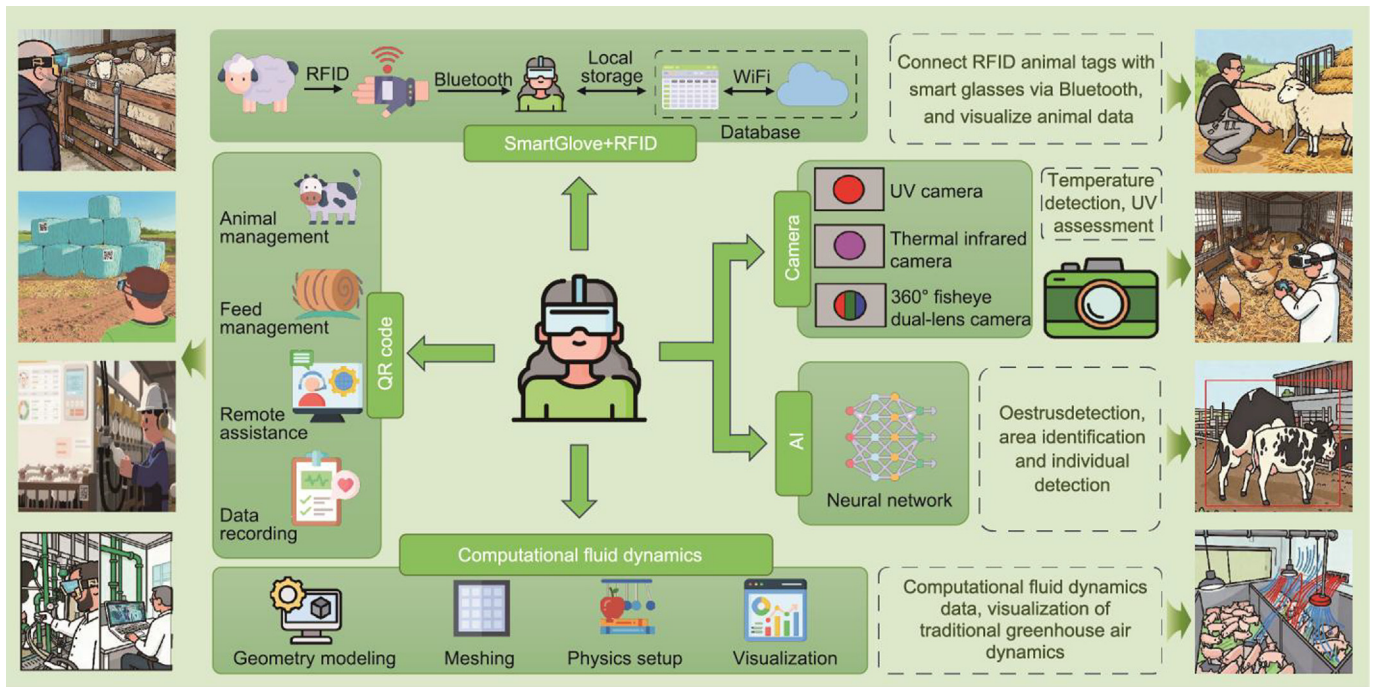
Furthermore, the usability of AR devices in livestock settings has been thoroughly investigated. Studies such as those by Caria et al. [153,154] have evaluated AR smart glasses from the perspectives of human-computer interaction performance and practical utility, confirming their significant potential as operational tools in livestock farming.

In the livestock industry, XR technology also plays a significant role in professional training through virtual simulation, effectively addressing challenges related to visualization and instruction caused by biosecurity restrictions and spatial isolation. For instance, Kim et al. [156] developed an immersive VR piglet housing simulator based on CFD simulation data. This simulator

Table 2

A comprehensive summary of applications in livestock and aquaculture management.

Application category	Application scenario	Target/object	Display device	XR type	Reference
Livestock farming	Individual identification	Dairy cow	Portable device	AR	[149]
Livestock farming	Individual identification & data management	Sheep	AR HMD	AR	[150]
Livestock farming	Equipment maintenance	Milking machine	AR HMD	AR	[151]
Livestock farming	Estrus detection & individual identification	Dairy cow	Portable device	AR	[152]
Livestock farming	Human-computer interaction	Livestock operation	AR HMD	AR	[153]
Livestock farming	Performance & usability evaluation	Dairy cow	AR HMD	AR	[154]
Livestock farming	Animal welfare management	Chicken	AR HMD	AR	[155]
Livestock farming	CFD	Piglet house	VR HMD	VR	[156]
Aquaculture	Full-process data support	Aquaculture system	AR HMD	AR	[157]
Aquaculture	Intelligent management	Prawn	AR HMD	AR	[158]
Aquaculture	Real-time growth monitoring	Abalone	Portable device	AR	[159]

**Fig. 7.** XR applications in agricultural management and monitoring system architectures: quick response (QR) code [153], SmartGlove + RFID [150], computational fluid dynamics [156], camera [155], and AI [152]. UV: ultraviolet.

visualizes ventilation and airflow conditions inside the pig house. Its primary utility lies in its ability not only to calculate the indoor microclimate according to external weather conditions and ventilation types but also to serve as an innovative educational tool, guiding farmers and advisors in scientifically designing and optimizing swine housing environments.

3.2.2. Aquaculture

In aquaculture, XR integration with IoT, CC, and AI supports environmental monitoring, growth observation, and intelligent decision support. This multi-technology approach is particularly suited to complex and dynamic aquatic conditions. Xi et al. [157] proposed an end-to-end sensor network + cloud platform + AR framework, where the cloud platform handles water-quality data collection, storage, analysis, and historical traceability, while the AR interface overlays farm-wide water-quality information and supports the rapid identification of abnormal ponds. This combined workflow improves farmers' decision-making efficiency and reduces their operational burden.

Further advancing this approach, Rahman et al. [158] developed a sensing-machine learning-AR system that uses sensors to collect

real-time data on key water parameters (e.g., dissolved oxygen, pH, and temperature) and applies machine learning models to predict pond conditions over the next 24 h. Risk warnings are then visually presented to farm managers via an AR HMD, enabling intelligent and proactive management in shrimp farming. In an initial on-farm trial using the farm's existing management practice as the baseline, the AR interactive interface helped managers reduce decision-making time by approximately 41% while achieving a decision accuracy rate of up to 89.2%. These results indicate the potential of AR-assisted decision support in aquaculture but should be interpreted within this specific evaluation setting. Napier and Lee [159] explored the application of XR technology at the individual organism level.

Combining mobile AR with object-detection technology, their system uses a smartphone camera to capture images of abalone, employs a deep learning model to analyze and measure growth parameters such as size in real time, and overlays the data visually onto the live animal through an AR interface. This enables real-time, non-contact monitoring of abalone growth status, effectively reducing the errors and costs associated with traditional manual measurement.

3.2.3. Summary and discussion

A core enabling framework emerges from XR research across livestock and aquaculture applications: state perception–intelligent decision-making–enhanced interaction. This framework describes a data workflow from the physical farming environment to digital management decisions and back to physical actions. In the state perception layer, AR headsets, smartphone cameras, RFID, multispectral sensors, and environmental sensors collect data on animal identity, physiological status, behavior, and environmental parameters. In the intelligent decision-making layer, cloud or edge AI models analyze these data to support individual identification, estrus detection, disease prediction, growth forecasting, and environmental control. The enhanced interaction layer then converts the outputs into intuitive visual symbols, such as virtual labels, highlighted boxes, alert icons, and holographic data panels, and overlays them on real or virtual environments to help managers interpret information and make decisions. By creating an intuitive data interface, this framework can reduce reliance on labor-intensive human observation in selected tasks and offers a promising route toward greater precision and efficiency in livestock and aquaculture management.

Nevertheless, the methodological maturity of XR applications in livestock and aquaculture remains limited. Many systems have been evaluated in laboratory settings, research farms, or small-scale experimental facilities, while long-term evidence under commercial production conditions is still scarce. In livestock farming, animal movement, occlusion, housing structures, and biosecurity constraints may reduce the stability of visual recognition and AR registration. In aquaculture, underwater visibility, water turbidity, sensor reliability, and the spatial separation between aquatic organisms and users further complicate real-time XR interaction. Therefore, although XR can improve data visibility and decision-making efficiency, the available evidence remains task-specific. For example, one AR-assisted shrimp farming study reported an approximately 41% reduction in decision-making time and a decision accuracy of up to 89.2% in an initial on-farm trial [158]; however, these results should be interpreted as field-based and preliminary. Further validation across different farms, operators, production cycles, and management conditions is still needed to assess long-term reliability, sustained accuracy, and cost-effectiveness in commercial-scale livestock and aquaculture operations.

3.3. XR-enhanced human–machine collaboration in agricultural machinery

The increasing diversity of agricultural machinery and the persistent decline in the agricultural labor force have made intelligent and automated farming solutions increasingly urgent [22,160]. Unlike highly structured industrial settings, however, agricultural environments are unstructured and demand flexible execution and intelligent decision-making. Fully autonomous and unmanned agricultural systems therefore remain a distant goal [161], whereas exclusive reliance on human labor entails high costs and low efficiency. Researchers have consequently explored an XR + agricultural machinery human–machine collaboration paradigm that combines human judgment with machine efficiency, through immersive interaction and virtual–real integration, XR can improve information exchange between operators and machinery, enhancing perceptual awareness and operational precision in onsite, remote, and virtual environments [162], as summarized in Table 3 [21,42,48,50,58,163–177]. The operating principles of XR in field control, remote control, and virtual simulation are illustrated in Fig. 8.

3.3.1. AR for enhanced situational awareness and guidance

AR technology aims to achieve the integration of multi-source information and intuitive spatial guidance, addressing issues such as information overload, high cognitive load, and the difficulty of managing multiple devices in traditional agricultural machinery cockpits. Its primary approach is to overlay critical real-time data and spatially registered virtual information on the work scene. This allows operators to access guidance without repeatedly shifting attention among conventional displays. The resulting integrated visual interface can improve decision-making efficiency, operational precision, and safety in complex agricultural environments.

These theoretical advantages have been supported by several empirical studies. For example, Jaiswal et al. [163] designed a heads-up display (HUD) unit for tractors based on a multibody model, which virtually projects key operational data—including engine speed, tire slip rate, bucket pose, and load—onto the windshield. This allows the driver to access instrument information without looking away, substantially improving both information accessibility and operational safety. In addition, Kaizu and Choi [164] developed an AR navigation system based on high-

Table 3

A comprehensive summary of applications in agricultural machinery.

Application Category	Application scenario	Target/object	Display device	XR type	Reference
Enhanced situational awareness and guidance	Design of tractor HUD	Tractor	HUD	AR	[163]
	Tractor navigation system	Tractor	Monitor	AR	[164]
	Supervision of multi-robot systems	Multirobot system	AR HMD	AR	[165]
Immersive teleoperation	Tractor navigation system	Tractor	AR HMD	AR	[166]
	Remote control	Agricultural robotic system	VR HMD	VR	[167]
	Remote control	Paddy transplanter	Monitor	VR/AR	[168]
	Remote control	Agricultural robotic system	VR HMD	VR	[21]
	Remote control	Agricultural robotic system	VR HMD	VR	[42]
	Remote control	Agricultural robotic system	VR HMD	VR	[169]
	Remote control	Agricultural robotic system	VR HMD	VR	[170]
	Remote control	Agricultural robotic system	VR HMD	VR	[171]
Virtual simulation	Agricultural machinery simulation	Tractors and operating machines	Monitor	VR	[48]
	Agricultural machinery simulation	Wheel tractor	Monitor	VR	[58]
	Agricultural machinery simulation	Tractor	Monitor	VR	[172]
	Agricultural machinery simulation	Harvester	Monitor + projector	VR	[173]
	Agricultural machinery simulation	Combine harvest	Monitor	VR	[174]
	Agricultural machinery simulation	Combine harvest	Monitor	VR	[175]
	Agricultural machinery simulation	Tractor	VR HMD	VR	[176]
	Agricultural drone simulation	Drone	VR HMD	VR	[177]
	Agricultural machinery simulation	Tractor	VR HMD	VR	[50]
	Agricultural machinery simulation	Tractor	VR HMD	VR	[50]

HUD: heads-up display.

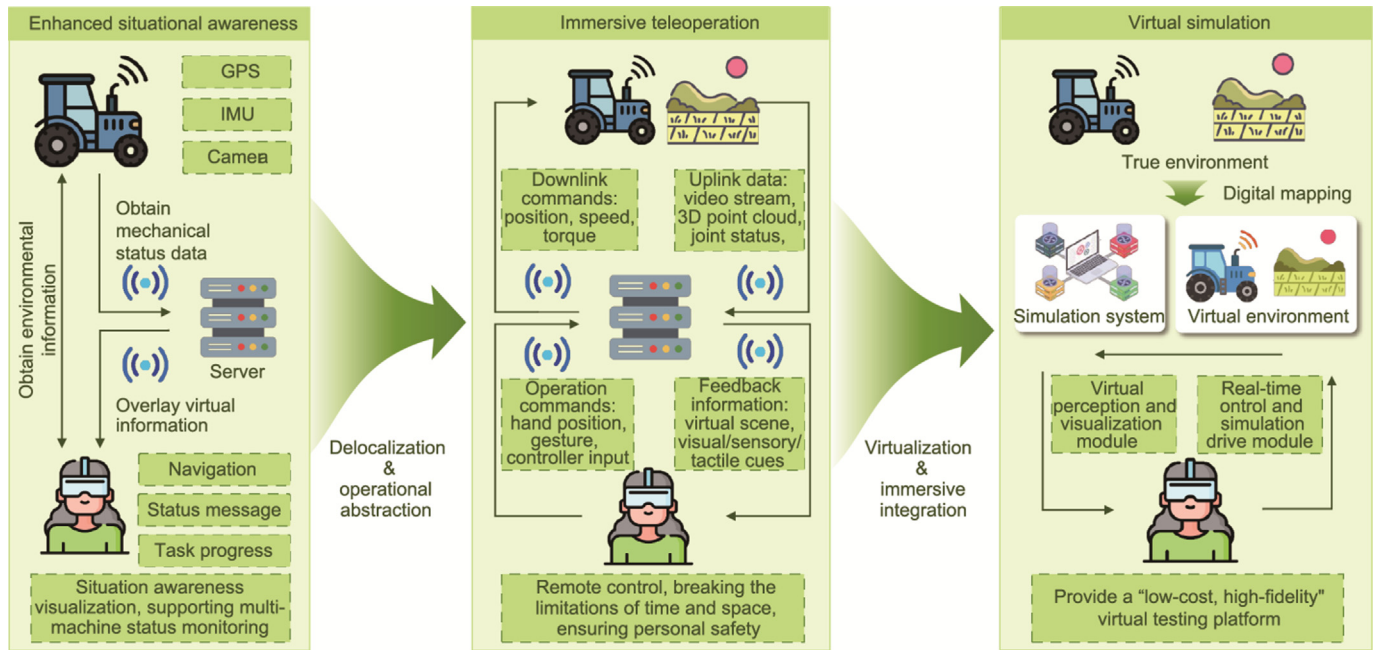


Fig. 8. Principles of XR applications in field control, remote control, and virtual simulation.

precision positioning technology. Using a virtual–real fusion approach, the system superimposes planned route lines directly into the operator's real-world view, effectively mitigating issues such as poor visibility during nighttime operations and the lack of intuitiveness in conventional navigation methods. Tests showed that the system achieves a positioning error of less than 3 cm and an end-to-end latency from image capture to display of under 10 ms, demonstrating excellent performance. Going a step further, Huuskonen and Oksanen [165] employed AR technology to construct a multi-robot monitoring system. Through an AR HMD, an operator can visually monitor the real-time status, speed, location, and other critical information of two autonomous agricultural machines simultaneously, enabling the efficient management and safe operation of machinery fleets and providing a reference solution for multi-machine coordination challenges.

3.3.2. VR for immersive teleoperation

In the domain of teleoperation and immersive control, researchers often favor VR technology over AR due to the former's ability to create a fully immersive virtual environment that isolates operators' senses from their physical surroundings. This allows for focused decision-making and control based on clear sensor data, significantly enhancing telepresence and immersion in remote operations. Abdulsalam and Aouf [167] developed a seamless teleoperation pipeline named VitRob Pipeline, which enables bidirectional, real-time data exchange—such as camera feeds, odometry, and velocity commands—between a VR headset and an agricultural robot. By presenting the environment virtually, the system offers an immersive experience that supports more informed remote decision-making, a capability validated through a fruit harvesting task. Lohan et al. [168] designed a remote-control system for a two-wheel rice transplanter. Operators use a VR headset to access an immersive control interface, manipulate the machine via VR hand controllers, and observe real-time camera feedback along with overlaid navigational path guidance. Udekwe and Seyyedhasani [21] integrated VR devices with SteamVR and the Unity 3D engine to develop a remote interaction system for agricultural robots. Using VR hand controllers, users can perform fruit-harvesting and leaf-inspection tasks. In a controlled laboratory

experiment using bionic tomatoes, the system exhibited a robot-plant interaction success rate of up to 90%, where “success” referred to harvesting the target fruit without damaging the plant or fruit. Because this result was obtained in a small task-specific setup without direct comparison with manual harvesting or a matched autonomous robotic baseline, it should be interpreted as feasibility evidence rather than general field performance. Liu et al. [42] constructed an immersive human–machine teleoperation framework that employs an unmanned aerial vehicle (UAV) to acquire real-time farmland data and construct a high-precision 3D digital map. This map is then rendered within a VR environment, and a closed-loop control mechanism is established between the physical machine and the virtual interface.

3.3.3. VR for virtual simulation

Leveraging VR's capacity for immersive virtual simulation, researchers have applied it to safety training and system evaluation in order to address long-standing challenges in agricultural machinery. The central approach is to reproduce machinery appearance, interfaces, controls, physical behavior, and operational workflows in a high-fidelity, risk-free environment, reducing dependence on costly physical equipment and exposure to operational hazards. Liu et al. [58] developed key components of a desktop VR driving simulator that closely reproduces wheeled-tractor operation and functionality. It provides novice drivers with a safe, standardized training environment while reducing training costs and risks. Faust et al. [172] systematically evaluated a high-fidelity tractor simulator in terms of appearance, interface, control, and sound, supporting its reliability for agricultural machinery safety research. Cutini et al. [48] designed a high-precision cooperative simulator for precision-agriculture machinery. It includes virtual tractors equipped with sprayers, seeders, and fertilizer spreaders, together with joysticks, control consoles, touchscreens, and multiple virtual-environment monitors, enabling complete workflows to be reproduced across field scenarios. Meusel et al. [173] used a high-fidelity combine-harvester simulator to evaluate two harvest-automation technologies: Advisor, which provides assistive prompts, and Director, which provides automatic control. Both received operator acceptance, but Director performed better.

The study also confirmed that high-fidelity simulators can evaluate agricultural automation technologies without the high cost, long duration, and risk of field tests.

3.3.4. Summary and discussion

An analysis of the application of XR in the agricultural machinery field reveals a common technology-empowerment framework of perception enhancement–control reconstruction–environment simulation. In the perception enhancement phase, AR is the primary technology, as it focuses on field operations. By overlaying information layers onto the real world, abstract machine data and spatial instructions are transformed into intuitive visual symbols, directly enhancing the operator's understanding of the current operational situation. The user primarily controls the equipment directly. Remote operations are mainly facilitated through VR technology. Sensors transmit onsite data to the data center, constructing an immersive virtual environment and establishing a bidirectional data link. In this case, the operator does not directly control the agricultural machinery but instead transmits control instructions to the server, realizing a shift from onsite operation to remote-presence operation and thus breaking geographical constraints. Environmental simulation, on the other hand, creates a high-fidelity, configurable virtual environment, digitizing complex and high-risk field operation processes to provide a safe, cost-effective, and repeatable digital sandbox for training, testing, and solution validation.

By reshaping the human–machine interaction interface, XR technology establishes a continuum from AR to VR in the complex human–machine system of agricultural machinery. This allows for the deep integration of human flexibility and judgment—which offer cognitive advantages—with the machine's precision and durability in the most suitable interaction modes, thus guiding the development of safe, efficient, and intelligent next-generation agricultural machinery systems.

The reviewed machinery studies also show that AR and VR differ substantially in their suitable operational scenarios and performance determinants. AR is more suitable for onsite operational guidance because it preserves the operator's direct perception of the machine and field environment, but its effectiveness critically depends on positioning accuracy, spatial registration, and outdoor display visibility. VR, in contrast, is better suited for teleoperation and virtual simulation, but it is highly sensitive to communication latency, environmental reconstruction fidelity, sensor completeness, and limited haptic or force feedback. These differences suggest that future evaluations should consider not only task accuracy or completion time but also safety, operator workload, latency tolerance, and transferability from virtual simulation to real machinery operation.

3.4. Education and training applications

Smart agriculture encompasses agricultural ecosystems and seeks intelligent, digital, and personalized production [178]. Although AI and related technologies are increasingly being integrated into practice, the effectiveness of advanced equipment ultimately depends on personnel who can operate it correctly, manage systems scientifically, and make accurate decisions [179]. Efficiently training farmers, managers, and new practitioners to operate precision-agriculture systems is therefore a central challenge [180]. By creating immersive, risk-free, and repeatable environments, XR can mitigate the resource, spatial, temporal, and safety constraints of conventional training, helping practitioners acquire smart-farming skills and indirectly supporting agricultural productivity. Numerous studies have begun to examine this role (Fig. 9 [64,181,182]).

For hands-on skill training, XR provides an immersive, risk-free environment for practicing high-risk and high-precision agricultural operations. Strousopoulos et al. [181] integrated gamification into agricultural VR education, using role-playing and task-based challenges to simulate swarm management and pest and disease control. This approach addresses the practical risks of conventional beekeeping training and its limited appeal among younger learners. Kee and Zhang [64] developed a digital experiential learning system that enables students to practice operations such as tree pruning repeatedly through VR devices, minimizing damage to real trees and reducing the carbon emissions associated with field visits.

For farm-system learning, XR can complement physical visits by providing immersive experiences at the farm scale. The 4D Virtual Farm platform [63] allows users to explore beef, lamb, and dairy farms virtually and can be integrated into learning management systems for remote education. Ario et al. [182] developed HoloFarm, which uses spatial audio to enhance immersion and reproduces the production cycle from soil preparation to harvest, enabling experimentation without wasting resources. Lasseur et al. [183] combined VR with biophysical models for grassland carbon-sequestration planning; users can design afforestation scenarios and examine real-time trade-offs among carbon sequestration, economic returns, and aesthetic value. Parikh et al. [184] created a peer-to-peer knowledge-sharing platform using detailed 3D farm models to facilitate expertise transfer between urban and rural farmers. Schütz et al. [185] evaluated virtual tours of pig farms, through which users learned about breeding and animal-welfare practices. Participants viewed this approach as useful for industry transparency and knowledge dissemination in areas where biosecurity limits physical visits. Table 4 [49,63,64,181–186] provides a comprehensive summary of XR applications in agricultural education and training.

Several systems discussed in Sections 3.1–3.3 also support training, including crop and environmental simulators [34,143,144], the piglet-housing simulator [156], and machinery simulators [48,58,172,173]. These systems remain classified by their primary application domains; Section 3.4 covers applications designed principally for education, training, or knowledge transfer.

3.5. Critical synthesis of current agricultural XR studies

Although the studies reviewed above collectively demonstrate the considerable potential of XR technologies in agriculture, the current evidence base remains heterogeneous and should be interpreted with caution. Reported improvements in operational efficiency, decision support, training effectiveness, and user engagement are derived from different XR modalities, display devices, task settings, performance indicators, and validation environments. Therefore, these findings are not always directly comparable, and technological promise must be distinguished from field-ready applicability.

A key inconsistency lies in the persistent gap between promising prototype-level results and limited large-scale deployment. Another limitation is that direct head-to-head comparisons of VR, AR, and MR for the same agricultural task remain limited in the reviewed literature. Their relative value should therefore be interpreted according to task requirements. For example, in agricultural machinery training and operation, VR is more suitable for risk-free simulation, teleoperation practice, and repeated workflow training, whereas AR is more suitable for onsite guidance, navigation, maintenance assistance, and real-time decision support because it preserves the operator's awareness of the physical environment. MR may support deeper spatial anchoring and collaborative interaction, but its agricultural use remains less mature. These

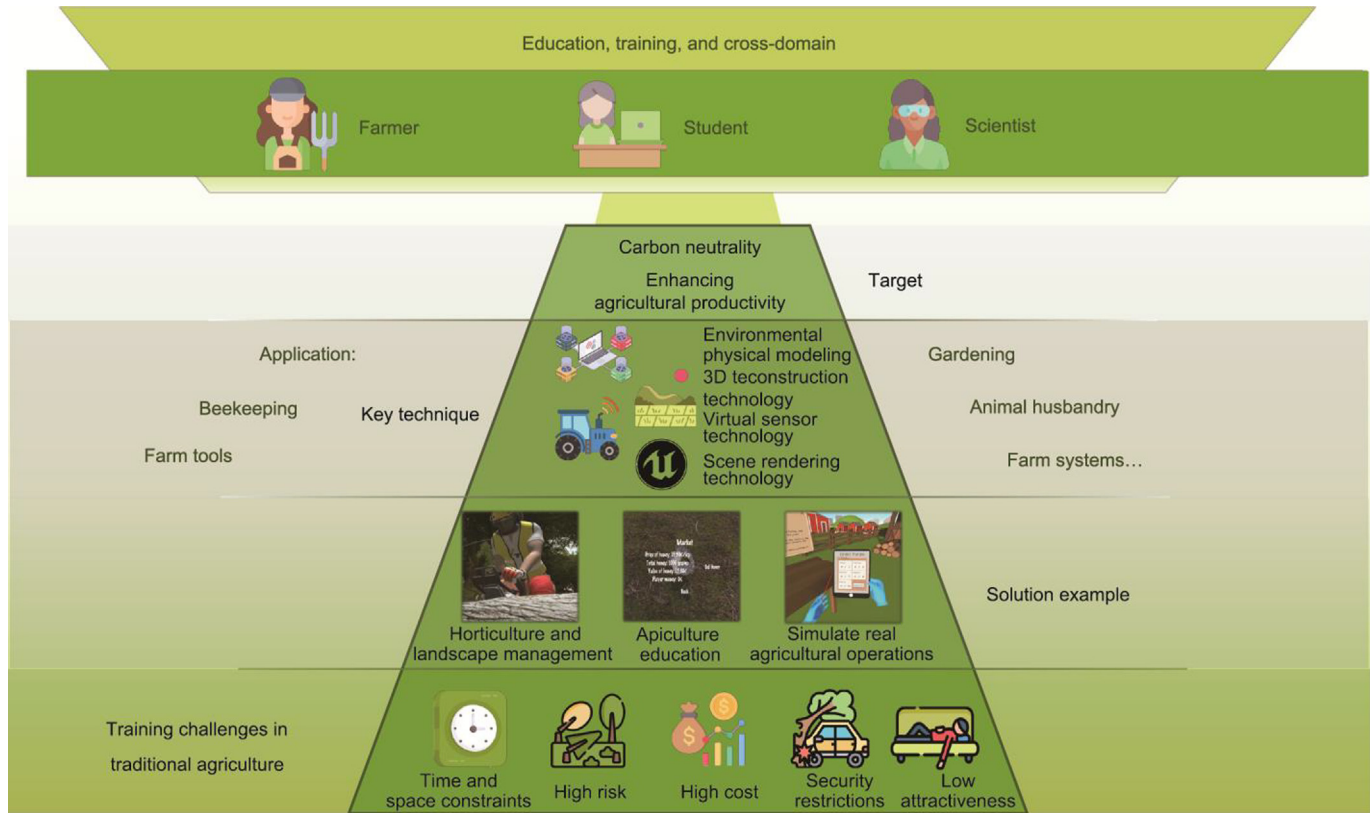


Fig. 9. XR-based education and training applications in agriculture: solution examples from Refs [64,181,182].

Table 4

A comprehensive summary of XR applications in agricultural education and training.

Application category	Application scenario	Target/object	Display device	XR type	Reference
Skill training and simulation	Apiculture education	Apiculture	VR HMD	VR	[181]
	Horticulture and landscape management	Trees	VR HMD	AR/VR	[64]
	Agricultural learning	Agricultural learning platform	VR HMD	VR	[182]
Farm-system learning and decision support	Farm tree planting management	Trees	VR HMD	VR	[183]
	Farm systems learning platform	Farm system	VR HMD/monitor/ portable device	VR	[63]
Public education and knowledge transfer	Agriculture educational curriculum	Veggies and fruit	portable device	AR	[49]
	Agricultural tools museum interaction system	Agricultural tool	VR HMD	VR	[186]
	Urban agriculture education	Virtual smart city	VR HMD	VR	[184]
	Pig husbandry education	Pig husbandry	VR HMD/portable device	VR	[185]

trade-offs indicate that technology selection should consider not only task performance but also usability, safety, user comfort, field awareness, and deployment complexity. More broadly, AR-based systems have shown potential for *in situ* visualization, crop diagnosis, livestock identification, and operational guidance, but their performance is strongly affected by outdoor illumination, occlusion, plant or animal movement, dust, humidity, registration accuracy, and display visibility. VR-based systems offer advantages in machinery simulation, teleoperation, digital twins, and training, yet their effectiveness depends heavily on model fidelity, communication latency, sensor feedback, and user comfort, making it difficult to generalize results from controlled settings to complex field operations. MR holds theoretical advantages for deeper virtual-physical interaction but remains less mature than AR and VR in agricultural applications, as it requires robust spatial mapping,

environmental understanding, and real-time interaction with deformable plants, moving animals, and irregular field structures.

Common methodological weaknesses further limit cross-study comparability. Many systems remain at the prototype or proof-of-concept stage, with validation conducted in laboratories, greenhouses, simulation platforms, or small-scale plots. Sample sizes are often limited, actual farmers or machinery operators are not always included, and evaluation metrics vary widely and lack standardization. Moreover, critical adoption factors such as long-term reliability, economic feasibility, maintenance requirements, device durability, user fatigue, and return on investment are rarely assessed. Few studies provide longitudinal evidence across seasons, crop growth stages, or varying farm conditions, leaving the stability of XR systems under cumulative environmental stress insufficiently understood.

Overall, current agricultural XR research has successfully demonstrated technological feasibility, but stronger evidence is still needed regarding field robustness, scalability, and practical adoption. Future work should move beyond isolated demonstrations toward standardized, task-specific, and longitudinal evaluation frameworks that clarify not only whether XR systems can function in agriculture but also when, where, for whom, and under what conditions they deliver reliable and economically meaningful value.

4. Challenges and future trends

XR combines virtual and real environments to varying degrees [187] and has been explored in crop cultivation, livestock management, aquaculture, and agricultural machinery. However, most applications remain at an early stage, with unresolved barriers to usability, environmental robustness, validation scale, and deployment [123,154,155,160].

Agricultural XR barriers are amplified by unstructured, less-controlled environments; deformable and seasonally changing biological targets; and task requirements that vary with weather, growth stage, farm scale, and operator experience. The barriers identified below are therefore interpreted in light of the reviewed evidence and these domain-specific conditions, rather than as generic XR limitations.

- **Device limitations:** Current XR devices still fall short of meeting the demanding requirements of outdoor agricultural environments. Mobile-phone-based solutions are limited by suboptimal display quality and the need to occupy the user's hands, greatly reducing practicality; monitor-based systems are even less portable. Although HMDs are currently among the most convenient form factors [188], they are still constrained by poor visibility under bright outdoor light, limited battery life, discomfort during prolonged use, and thermal performance degradation under direct sunlight [189]. These limitations are particularly evident in the AR-based crop and machinery applications reviewed in Sections 3.1 and 3.3, where HMDs or portable devices are expected to support *in situ* visualization during harvesting, pest diagnosis, soil sampling, or machinery navigation. Unlike indoor medical or manufacturing XR scenarios, agricultural users often work under strong sunlight, dust, rain, wind-induced plant movement, and large temperature variations. These conditions directly affect display visibility, camera-based recognition, SLAM-based tracking, spatial registration, battery life, and thermal stability, making outdoor robustness a more critical requirement for agricultural XR than for many other XR application domains.
- **Software and ecosystem integration challenges:** The effective deployment of XR technology relies on deep integration with CV, AI, IoT, and edge computing [190]. While this allows XR to benefit from advances in these fields, it also constrains XR within their current development. Persistent issues exist such as inconsistent protocol interfaces, poor cross-platform compatibility, and weak data interoperability, leading to low adaptation efficiency [13,191,192]. The reviewed studies also show that software and algorithmic limitations are closely tied to agricultural object variability. For example, crop disease diagnosis [51,129,130], strawberry ripeness detection [120,122], grape berry thinning [121], rice planthopper counting [131], and leaf-miner damage assessment [132] all depend on reliable CV and AI models. However, these models are often developed for specific crops, pests, diseases, or controlled imaging conditions. In real farms, plant morphology changes across growth stages, leaves and fruits are frequently occluded, and crop rows may present repetitive visual patterns, all of which complicate recognition and registration. Such situations differ from many industrial XR applications, where target objects are rigid, standardized, and spatially predictable. Therefore, agricultural XR requires not only general XR middleware but also crop- and task-adapted perception models, expert knowledge systems, and interoperable data interfaces.
- **Economic cost and infrastructure barriers:** The complexity and unstructured nature of agricultural production make it difficult to develop universal XR solutions. In many remote areas, insufficient network coverage and inadequate computing infrastructure further raise the barriers to deployment, particularly for VR teleoperation, digital twin systems, and cloud-assisted recognition, which require low-latency communication, stable bidirectional data transmission, and substantial computing resources. Economic barriers are also more pronounced in agriculture than in many institutional XR application domains. While hospitals, factories, or universities may support the centralized deployment and maintenance of XR systems, agricultural production is often distributed across small and medium-sized farms with limited investment capacity [14]. The reviewed systems frequently combine XR devices, AI models, IoT sensors, cloud/edge computing, and customized software, which increases not only initial deployment costs but also maintenance and technical support requirements. For small and medium-sized farms, the return on investment is therefore likely to depend on whether XR can be applied to recurrent, high-value, or safety-critical tasks where gains in labor efficiency, input-use precision, training effectiveness, or downtime reduction can offset deployment and maintenance costs. This partly explains why many studies remain at the prototype stage despite promising experimental results.
- **Methodological limitations and evidence gaps:** Current agricultural XR research also faces several methodological limitations. Many studies emphasize system development and feasibility demonstration rather than rigorous comparative evaluation. Validation settings vary widely, including laboratory simulations, greenhouse experiments, and small-scale field tests, making cross-study comparison difficult. Evaluation indicators are also inconsistent, ranging from recognition accuracy and latency to task completion time, immersion, usability, or educational outcomes. Furthermore, limited participant numbers and insufficient involvement of actual farmers or machinery operators may reduce the practical relevance of usability conclusions. These evidence gaps partly explain why many promising XR prototypes have not yet been translated into widespread agricultural adoption, and they indicate the need for standardized benchmarks, longitudinal field trials, and user-centered evaluation protocols in future research.
- **Data privacy, ethical governance, and regulatory risks:** Data concerning field conditions, crop growth, livestock status, farming scale, machinery operation, and worker activities are not only large in volume but also commercially sensitive and, in some contexts, relevant to food security [193–195]. This concern is particularly relevant to XR systems integrating IoT sensors, cloud platforms, livestock identification, aquaculture monitoring, farm-scale digital twins, and remote machinery operation, because these applications require continuous collection, transmission, and visualization of production data. Beyond data security, agricultural XR raises broader ethical and governance issues. XR devices, cameras, positioning systems, and operation logs may enable continuous monitoring of farm workers and machinery operators [15]. Therefore, informed consent, transparent data-use purposes, clear retention rules, and safeguards against inappropriate surveillance are essential. Data ownership and sovereignty are also critical in smallholder and cooperative farming contexts, where field

data, crop health information, yield predictions, and operational records may become valuable economic assets. Farmers should retain meaningful control over how these data are accessed, shared, commercialized, or transferred to technology providers. In addition, XR-assisted autonomous machinery and remote operation systems require clear regulatory and accountability frameworks, including human oversight, traceable operation logs, emergency-stop mechanisms, and compliance with relevant data protection, labor, occupational safety, and autonomous machinery regulations.

- **Lack of long-term validation in real agricultural environments:** Most current XR solutions are tested primarily in controlled laboratories, greenhouses, or short-term demonstrations, preventing a systematic assessment of their cumulative performance under complex and variable field conditions [196]. This limitation is particularly problematic in agriculture because production systems are inherently seasonal and biologically dynamic. A system that performs well during a short-term greenhouse or laboratory test may not remain reliable across different crop growth stages, weather conditions, production cycles, or animal behaviors. For example, recognition and registration accuracy may decline as canopy structures change, animals move unpredictably, sensors drift, or hardware is exposed to dust, humidity, and heat over time. As a result, critical issues such as gradual sensor calibration drift, long-term algorithmic stability, and hardware durability remain insufficiently investigated. Without robust longitudinal field data, users may reasonably question the reliability and practical value of XR systems, hindering their widespread adoption and scalable integration [197].
- **User adaptation and social acceptance challenges:** The current agricultural workforce is still predominantly composed of middle-aged and older individuals, who tend to rely on accumulated practical experience and may exhibit significant resistance to adopting emerging technologies such as XR. Many target users of agricultural XR systems are farmers, livestock managers, aquaculture operators, or machinery drivers, whose age, digital literacy, and working habits may differ substantially from those of participants in laboratory usability studies. Moreover, agricultural work often requires physical mobility, tool handling, and attention to safety-critical surroundings. Therefore, device comfort, hands-free interaction, cognitive workload, and trust in AI-generated recommendations are not secondary usability issues but decisive factors for real-world adoption. Coupled with the current immaturity of the technology and the risk of inaccurate AI/CV outputs, poor user adaptation may reduce user trust and hinder the long-term integration of XR in agriculture.

Looking ahead, XR could become an enabling technology for agricultural production, although substantial time, infrastructure, and investment will be required. Its value lies in convergence with AI, image processing, digital twins, and related technologies rather than in isolated deployment. Interdisciplinary collaboration is equally important: As previous studies indicate [193,198], agricultural XR cannot be developed by computer scientists alone; agronomists, engineers, and farmers must help align systems with production needs. Ethics-by-design and governance-by-design principles should also ensure transparency, accountability, informed consent, data sovereignty, and safety from the outset. Wider adoption will require lower hardware costs and more usable software, supported by deployment at scale, open-source ecosystems, and user-centered design. Affordable systems could create a cycle of technical improvement, cost reduction, and broader adoption. XR may also advance agricultural sustainability [34] by combining information processing and expert systems for data visualization and decision support [199], improving water and fertilizer use [52,200], reducing chemical inputs, mitigating environ-

mental pollution [201], and lowering the carbon footprint of production [202]. Its immersive capabilities can further support public education, narrow urban–rural knowledge gaps, and strengthen public understanding of and trust in modern agriculture.

5. Conclusions

This review has elucidated the potential of XR technologies in bridging the critical gap between the digital and physical realms of modern agriculture and highlighted XR's key role in addressing the complex challenges presented by agriculture. As a technology that connects the digital and physical worlds, XR not only changes the way interactions occur in agriculture but also enhances the information access and decision-making capabilities of agricultural practitioners and production systems.

The review first emphasizes the urgent need for technological innovation in agriculture, leading to the core value of XR technology: It addresses the “last-mile” human–machine interaction issue in agricultural work. Specifically, XR can present complex AI- and IoT-derived information through intuitive interfaces, helping front-line workers make faster and better-informed decisions. In this way, XR can reduce some interaction barriers in conventional digital systems. Our analysis suggests that this value is beginning to emerge through several application patterns across the agricultural spectrum. In crop management, XR has been used to construct task-specific perception–analysis–visualization–guidance workflows, from real-time operational guidance to digital twin systems supporting remote farm monitoring and management strategy simulation, increasing precision and efficiency. For livestock and aquaculture, it enables precise individual animal management through identification and welfare monitoring. In agricultural machinery, XR improves onsite navigation and serves as a useful tool for remote control and skill training via high-fidelity digital sandboxes. Finally, in education, it overcomes the inherent resource and safety limitations of traditional training models. Collectively, these applications facilitate a gradual transition from experience-based to data-informed agricultural operations.

However, the path to widespread adoption is fraught with significant obstacles. These include the technical immaturity of devices in harsh outdoor environments, economic costs and infrastructure barriers, data privacy concerns, socioeconomic challenges, a lack of long-term validation in real agricultural settings, and adoption barriers among aging users. Addressing these challenges requires concerted interdisciplinary efforts that move beyond incremental improvements. Future priorities must include developing ruggedized, cost-effective hardware with intuitive user interfaces; establishing open, interoperable software platforms to foster an innovation ecosystem; and creating robust, long-term validation frameworks to build trust and demonstrate tangible return on investment in real-world environments.

As technological limitations are overcome and economic costs decrease, XR may evolve from a promising tool into a practical component of sustainable agriculture if current technical, economic, and validation barriers are addressed. Its capabilities in optimizing resource utilization, minimizing environmental impact, and increasing resilience position it to contribute to addressing food-security and agricultural-sustainability challenges. This perspective is also consistent with broader sustainability discussions in agriculture, including the distinction between production efficiency and eco-efficiency [203], trust and transparency in organic and sustainable food systems [204], and the influence of energy–agriculture market linkages on agricultural transformation [205]. We therefore urge collaboration among researchers, engineers, policymakers, and farmers to construct a comprehensive XR

ecosystem for agriculture. The ultimate objective is clear: to leverage this powerful technology not merely for incremental gains but to support the ongoing transformation of agricultural production toward greater productivity, intelligence, and sustainability.

CRediT authorship contribution statement

Tao Liu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Zhonghua Miao:** Writing – review & editing, Writing – original draft, Supervision. **Simon X. Yang:** Visualization, Data curation. **Jun Zhou:** Validation, Supervision. **Liang Gong:** Validation, Data curation. **Baohua Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Chengliang Liu:** Writing – review & editing, Supervision, Methodology, Investigation. **Chunjiang Zhao:** Writing – review & editing, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Abbasi R, Martinez P, Ahmad R. The digitization of agricultural industry—a systematic literature review on agriculture 4.0. *Smart Agric Technol* 2022;2:100042.
- [2] Abu Samah A, Shaffril HAM, Fadzil MF, Ahmad N, Idris K. A systematic review on adaptation practices in aquaculture towards climate change impacts. *Sustainability* 2021;13(20):11410.
- [3] Raza A, Razzaq A, Mehmood SS, Zou X, Zhang X, Lv Y, et al. Impact of climate change on crops adaptation and strategies to tackle its outcome: a review. *Plants* 2019;8(2):34.
- [4] Gupta D, Gujre N, Singha S, Mitra S. Role of existing and emerging technologies in advancing climate-smart agriculture through modeling: a review. *Ecol Inform* 2022;71:101805.
- [5] Porter JR, Challinor AJ, Henriksen CB, Howden SM, Martre P, Smith P. Invited review: intergovernmental panel on climate change, agriculture, and food—a case of shifting cultivation and history. *Glob Change Biol* 2019;25(8):2518–29.
- [6] Quinteiro P, Ridoutt BG, Arroja L, Dias AC. Identification of methodological challenges remaining in the assessment of a water scarcity footprint: a review. *Int J Life Cycle Assess* 2018;23(1):164–80.
- [7] Buheji M. Foresighting the transformation requirements toward “aging economy”. *Int J Manag* 2021;12(9):123–35.
- [8] Zhang Z, Ding J, Zhang K, Deng X. Impact of aging on farmland abandonment: evidence from rural China. *Land* 2025;14(2):393.
- [9] Bochtis D, Benos L, Lampridi M, Marinoudi V, Pearson S, Sørensen CG. Agricultural workforce crisis in light of the COVID-19 pandemic. *Sustainability* 2020;12(19):8212.
- [10] Bono A, Guaragnella C, D'Orazio T. A perspective analysis of imaging-based monitoring systems in precision viticulture. *Artif Intell Agric* 2026;16(1):62–84.
- [11] Pandey DK, Mishra R. Towards sustainable agriculture: harnessing AI for global food security. *Artif Intell Agric* 2024;12:72–84.
- [12] Abbasi R, Martinez P, Ahmad R. An ontology model to represent aquaponics 4.0 system's knowledge. *Inf Process Agric* 2022;9(4):514–32.
- [13] Feng C, Sun C, Xu B, Luo N, Liu H. Agricultural metaverse: key technologies, application scenarios, challenges and prospects. *Smart Agric* 2022;4:126.
- [14] López-Cabarcos MÁ, Piñeiro-Chousa J. Illusion or reality? Building a metaverse community focused on value creation in the agricultural sector. *Int J Inf Manag* 2024;77:102782.
- [15] Bigonah M, Jamshidi F, Pant A, Poudel S, Nallapareddy SR, Langroudi AC, et al. A systematic review of extended reality (XR) technologies in agriculture and related sectors (2022–2024). *IEEE Access* 2025;13:49721–34.
- [16] Anastasiou E, Balafoutis AT, Fountas S. Applications of extended reality (XR) in agriculture, livestock farming, and aquaculture: a review. *Smart Agric Technol* 2023;3:100105.
- [17] Jones D, Ghasemi S, Gračanin D, Azab M. Privacy, safety, and security in extended reality: user experience challenges. In: *Lecture Notes in Computer Science-2023*, vol. 14045. Cham: Springer Science and Business Media Deutschland GmbH; 2023. p. 511–28.
- [18] Sun T, Yu Y, Zheng Q, Zheng C. Extended reality: enhancing human-centered capabilities for human–cyber–physical systems. *IFAC Pap* 2024;58:368–73.
- [19] Wang X, Shen L, Lee LH. Towards massive interaction with generalist robotics: a systematic review of XR-enabled remote human–robot interaction systems. *ACM Comput Surv* 2025;57(11):1–37.
- [20] Kumari S, Raghuram P, Venkatesh VG, Shi Y. Future perspectives on progressive farming with adoption of virtual reality technology for sustainable quality in agriculture. *TQM J* 2022;34(2):250–79.
- [21] Udekwe D, Seyyedhasani H. Human robot interaction for agricultural teleoperation, using virtual reality: a feasibility study. *Comput Electron Agric* 2025;228:109702.
- [22] Ronaghi M, Ronaghi MH. Investigating the impact of economic, political, and social factors on augmented reality technology acceptance in agriculture. *Technol Soc* 2021;67:101739.
- [23] Rokhsarilemi S, Sadeghi-Niaraki A, Choi SM. A review on mixed reality: current trends, challenges and prospects. *Appl Sci* 2020;10(2):636.
- [24] Sara G, Todde G, Pinna D, Caria M. Investigating the intention to use augmented reality technologies in agriculture. *Comput Electron Agric* 2024;224:109252.
- [25] Morimoto T, Kobayashi T, Hirata H, Otani, Sugimoto M, Tsukamoto, et al. XR (extended reality: virtual reality, augmented reality, mixed reality) technology in spine medicine. *J Clin Med* 2022;11(2):470.
- [26] Tang YM, Chau KY, Kwok APK, Zhu T, Ma X. A systematic review of immersive technology applications for medical practice and education. *Educ Res Rev* 2022;35:100429.
- [27] Koohang A, Nord JH, Ooi KB, Tan GWH, Al-Emran M, Aw ECX, et al. Shaping the metaverse into reality: a holistic multidisciplinary understanding of opportunities, challenges, and avenues for future investigation. *J Comput Inf Syst* 2023;63(3):735–65.
- [28] Lampropoulos G, Kinshuk. Virtual reality and gamification in education: a systematic review. *Educ Technol Res Dev* 2024;72(3):1691–785.
- [29] Purcell W, Neubauer T. Digital twins in agriculture: a state-of-the-art review. *Smart Agric Technol* 2023;3:100094.
- [30] Pyliaianidis C, Osinga S, Athanasiadis IN. Introducing digital twins to agriculture. *Comput Electron Agric* 2021;184:105942.
- [31] Hasani MF, Ario MK, Balqis K, Carment M, Christian L. Virtual reality for farming education and simulation. In: *Proceedings of the 2025 International Seminar on Intelligent Technology and Its Applications*; 2025 Jul 23–25; Surabaya, Indonesia. New York City: IEEE; 2025. p. 531–6.
- [32] Milgram F, Kishino F. Taxonomy of mixed reality visual displays. *IEICE Trans Inf Syst* 1994;E77-D:1321–9.
- [33] Flavián C, Ibáñez-Sánchez S, Orús C. The impact of virtual, augmented and mixed reality technologies on the customer experience. *J Bus Res* 2019;100:547–60.
- [34] Spyrou O, Ariza-Sentís M, Vélez S. Enhancing education in agriculture via XR-based digital twins. *Appl Syst Innov* 2025;8(2):38.
- [35] Har LL, Rashid UK, Chuan LT, Xia LY. Revolution of retail industry: from perspective of retail 1.0 to 4.0. *Procedia Comput Sci* 2022;200:1615–25.
- [36] Lee LK, Wei X, Chui KT, Cheung SKS, Wang FL, Fung YC, et al. A systematic review of the design of serious games for innovative learning. *Electronics* 2024;13(5):890.
- [37] Ferreira JMM, Qureshi ZI. Use of XR technologies to bridge the gap between higher education and continuing education. In: *Proceedings of the 2020 IEEE Global Engineering Education Conference*; 2020 Apr 27–30; Porto, Portugal. New York City: IEEE; 2020. p. 913–8.
- [38] Han J, Seo S, Howlett EA. Can augmented reality (AR) enhance how restaurants present farm-to-table information? *Int J Hospit Manag* 2026;132:104340.
- [39] Andrews C, Southworth MK, Silva JNA, Silva JR. Extended reality in medical practice. *Curr Treat Options Cardiovasc Med* 2019;21(4):18.
- [40] Spyrou O, Hurst W, Verdouw C. Virtual reality-based digital twins. *Big Data Cogn Comput* 2023;7(2):95.
- [41] González de Cosío Barrón A, Gonzalez Almaguer C, Berglund A, Apraiz I, Saavedra V, Peñalva J, et al. Immersive learning in agriculture: XR design of robotic milk production processes. In: *Proceedings of the 2024 International Conference on Engineering Production Design Education*; 2024 Sep 4–6; Birmingham, UK. Glasgow: The Design Society; 2024. p. 575–80.
- [42] Liu T, Zhang B, Tan Q, Zhou J, Yu S, Zhu Q, et al. Immersive human–machine teleoperation framework for precision agriculture. *Comput Electron Agric* 2024;226:109444.
- [43] Capece N, Manfredi G, Gilio G, Erra U, Toscano F, Fiorentino C, et al. An educational approach for mixed reality visualization of agro-meteorological parameters. In: *Proceedings of the 2023 IEEE International Conference on Metrology, Extended Reality, Artificial Intelligence, Neural Engineering*; 2023 Sep 11–13; Milano, Italy. New York City: IEEE; 2023. p. 46–51.
- [44] Gong P, Cao Y, Zhang J, Shi Y, Fu Z. AR-enabled design strategies for future agricultural cultural heritage experiences. In: *Cross-cultural design*. Cham: Springer Nature Switzerland; 2025. p. 374–92.

- [45] Xing Y, Cao Q. Research on the application of interactive design in online farmland. In: Design, user experience, usability. Cham: Springer Nature Switzerland; 2025. p. 258–70.
- [46] Sutherland IE. The ultimate display. *SciSpace Pap* 1965;2:506–8.
- [47] Daassi M, Debbabi S. Intention to reuse AR-based apps: the combined role of the sense of immersion, product presence and perceived realism. *Inf Manage* 2021;58(4):103453.
- [48] Cutini M, Bisaglia C, Brambilla M, Bragaglio A, Pallottino F, Assirelli A, et al. A co-simulation virtual reality machinery simulator for advanced precision agriculture applications. *Agriculture* 2023;13(8):1603.
- [49] AGV. Cultivating a green future: augmented reality veggie and fruit explorer app empowering Indian curriculum with agricultural knowledge. In: Proceedings of the 2023 International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering; 2023 Nov 01–02; Chennai, India. New York City: IEEE; 2023. p. 1–8.
- [50] Kanivets I, Kanivets O, Gorda TM. VR simulator for studying the structure of tractors. In: Proceedings of the 2nd Workshop on Digital Transformation of Education, co-located with the 18th International Conference on ICT in Education, Research, and Industrial Applications (ICTERI 2023); 2023 Sep 18–23; IvanoFrankivsk, Ukraine. Bonn: CEUR Workshop Proceedings; 2023. p. 125–41.
- [51] Balahadia F, Billones R, Valdez D. Development of an augmented reality framework for corn plant disease identification. *AIP Conf Proc* 2025;3287:030006.
- [52] Prashanth JS, Moparthi NR, Krishna GB, Prasad AVK, Sravankumar B, Rao PR. MPCsAR-AHH: a hybrid deep learning model for real-time detection of cassava leaf diseases. *Comput Electr Eng* 2024;119:109628.
- [53] Prabhu BVB, ARIA: augmented reality and artificial intelligence enabled mobile application for yield and grade prediction of tomato crops. *Procedia Comput Sci* 2024;235:2693.
- [54] Monica R, Rizzini DL, Aleotti J. Adaptive complementary filter for hybrid inside-out HMD tracking. *IEEE Trans Vis Comput Graph* 2025;31(2):1598–612.
- [55] Ahuja K, Shen V, Fang CM, Riopelle N, Kong A, Harrison C. ControllerPose: inside-out body capture with VR controller cameras. In: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems; 2022 Apr 29–May 05; New Orleans, LA, USA. New York City: ACM; 2022. p. 1–13.
- [56] Jiang X, Zhu L, Liu J, Song A. A SLAM-based 6DoF controller with smooth auto-calibration. *Vis Comput* 2023;39(9):3873–86.
- [57] Hurst W, Mendoza FR, Tekinerdogan B. Augmented reality in precision farming. *Smart Cities* 2021;4(4):1454–68.
- [58] Liu C, Chang C, Hu J. Development of key components of wheel tractor driving simulator based on desktop virtual reality. *IOP Conf Ser Earth Environ Sci* 2020;474(3):032019.
- [59] Akturk S, Valentine J, Ahmad J, Jagersand M. Immersive human-in-the-loop control. In: Proceedings of the 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems; 2024 Oct 14–18; Abu Dhabi, United Arab Emirates. New York City: IEEE; 2024. p. 1217–82.
- [60] McAnally K, Wallis G. Visual-haptic integration, action and embodiment in virtual reality. *Psychol Res* 2022;86(6):1847–57.
- [61] Fani S, Ciotti S, Catalano MG, Grioli G, Tognetti A, Valenza G, et al. Simplifying telerobotics: wearability and teleimpedance improves human–robot interactions. *IEEE Robot Autom Mag* 2018;25(1):77–88.
- [62] Lohan SK, Prakash C, Lohan N, Kansal S, Karkee M. State-of-the-art in real-time virtual interfaces for tractors. *Comput Electron Agric* 2025;231:109947.
- [63] Hallein E, Shallcross D, Dalveen J, Celi P, Jacobson C, et al. Collaborative development of a farm systems learning platform. *J Vet Med Educ* 2025;52(1):8–16.
- [64] Kee T, Zhang H. Digital experiential learning for sustainable horticulture and landscape management education. *Sustainability* 2022;14(15):9116.
- [65] Kong A, Feng Z. Advancing VR edutainment design in blended learning. *Comput Educ X Real* 2024;5:100078.
- [66] Zulkarnain AHB, Gere A. Virtual reality sensory analysis approaches for sustainable food production. *Appl Food Res* 2025;5:100780.
- [67] Sun D, Kiselev A, Liao Q, Stoyanov T, Loufi A. A new mixed-reality-based teleoperation system. *IEEE Trans Hum Mach Syst* 2020;50(1):55–67.
- [68] Ulagammai M, Moorthy RN. An augmented reality based intelligent precision agriculture. In: In: Proceedings of the 2023 7th International Conference on Trends Electronic Information; 2023 Apr 11–13; Tirunelveli, India. New York City: IEEE; 2023. p. 303–7.
- [69] Syed TA, Siddiqui MS, Abdullah HB, Jan S, Namoun A, Alzahrani A, et al. In-depth review of augmented reality: tracking technologies, development tools, AR displays, collaborative AR, and security concerns. *Sensors* 2023;23(1):146.
- [70] Xia X, Guan FY, Cai Y, Magnenat TN. Challenges and advancements for AR optical see-through near-eye displays: a review. *Front Virtual Real* 2022;3:838237.
- [71] Kaimaris D, Roustanis T, Karolos IA, Klimantakis K, Patias P. Aligning positioning and navigation of smart mobile devices to AR/VR environments. *Int Arch Photogramm Remote Sens Spat Inf Sci* 2021;XLVI-M-1:347–53.
- [72] Sheng X, Mao S, Yan Y, Yang X. Review on SLAM algorithms for augmented reality. *Displays* 2024;84:102806.
- [73] Ke S, Xiang F, Zhang Z, Zuo Y. An enhanced interaction framework based on VR, AR and MR. *Procedia CIRP* 2019;83:753–8.
- [74] Vasarainen M, Paavola S, Vetoshkina L. A systematic literature review on extended reality. *Int J Virtual Real* 2021;21(2):1–28.
- [75] Parveau M, Adda M. 3iVClass: a new classification method for virtual, augmented and mixed realities. *Procedia Comput Sci* 2018;141:263–70.
- [76] Choi MH, Han W, Min K, Min D, Shin KS, et al. Recent applications of optical elements in augmented and virtual reality displays. *ACS Appl Opt Mater* 2024;2(7):1247–68.
- [77] Biercewicz K. Using mixed reality for green marketing strategies. *Procedia Comput Sci* 2024;246:5488–97.
- [78] Li CH, Chan OLK, Chow YT, Zhang X, Tong PS, Li SP, et al. Evaluating the effectiveness of digital content marketing under mixed reality training platform. *Front Psychol* 2022;13:881019.
- [79] Yang F, Miang GYVR. VR and MR technology for safety management education. *Saf Sci* 2022;148:105645.
- [80] Zhang X, Bai X, Zhang S, He W, Wang P, Wang Z, et al. Real-time 3D video-based MR remote collaboration. *Int J Adv Manuf Technol* 2022;121(11–12):7697.
- [81] Baron AM, Gul LF. A survey of computer vision techniques and technologies for interactive media design in extended reality environments. *Int J Hum Comput Interact* 2025.
- [82] Ghazal S, Munir A, Qureshi WS. Computer vision in smart agriculture and precision farming. *Artif Intell Agric* 2024;13:64–83.
- [83] Dhakal A, Ran X, Wang Y, Chen J, Ramakrishnan KK. SLAM-share: visual simultaneous localization and mapping. In: Proceedings of the 18th International Conference on Emerging Networking Experiments and Technologies; 2022 Dec 6–9; Roma, Italy. New York City: ACM; 2022. p. 293–306.
- [84] Angrisani L, D'Arco M, De Benedetto E, Duraccio L, Lo Regio F, Sansone M, et al. Performance measurement of gesture-based human–machine interfaces within extended reality head-mounted displays. *Sensors* 2025;25(9):2831.
- [85] Plopski A, Hirzle T, Norouzi N, Qian L, Bruder G, Langlotz T. The eye in extended reality: a survey on gaze interaction and eye tracking. *ACM Comput Surv* 2022;55:1–39.
- [86] Hu Y, Xu J, Gou Z, Cui D. Recognizing and localizing chicken behaviors in videos based on spatiotemporal feature learning. *Artif Intell Agric* 2025;15(4):816–28.
- [87] Nasir MF, Fuentes A, Han S, Liu J, Jeong Y, Yoon S, et al. Multi-camera fusion and bird-eye view location mapping for deep learning-based cattle behavior monitoring. *Artif Intell Agric* 2025;15(4):724–43.
- [88] Wu N, Gao P, Wu J, Zhao Y, Xu X, Zhang C, et al. Rapid detection and visualization of physiological signatures in cotton leaves under verticillium wilt stress. *Artif Intell Agric* 2025;15(4):757–69.
- [89] Yang Y, Wang X, Zhang F, Wu Z, Wang Y, Liu Y, et al. MSNet: a multispectral-image driven rapeseed canopy instance segmentation network. *Artif Intell Agric* 2025;15(4):642–58.
- [90] Chakrabarty S, Deb CK, Marwaha S, Haque MA, Kamil D, Bheemanahalli R, et al. Application of artificial intelligence in insect pest identification—a review. *Artif Intell Agric* 2026;16(1):44–61.
- [91] Kaya Y, Gürsoy E. A novel multi-head CNN design to identify plant diseases using the fusion of RGB images. *Ecol Inform* 2023;75:101998.
- [92] Maitra A, Damle M. Revolutionizing plant health management with technological digital transformation to enhance disease control. In: 2024 International Conference on Innovative Technology (INOCON 2024); 2024 Mar 01–03; Bangalore, India. New York City: IEEE; 2024. p. 1–8.
- [93] Shoaib M, Shah B, El-Sappagh S, Ali A, Ullah A, Alenezi F, et al. An advanced deep learning models-based plant disease detection: a review of recent research. *Front Plant Sci* 2023;14:1158933.
- [94] Das A, Pathan F, Jim JR, Kabir M, Mridha MF. Deep learning-based classification, detection, and segmentation of tomato leaf diseases. *Artif Intell Agric* 2025;15(2):192–220.
- [95] Hassan SI, Alam MM, Zia MI, Rashid M, Illahi U, Su'ud MM. Rice crop counting using aerial imagery and GIS for the assessment of soil health to increase crop yield. *Sensors* 2022;22(21):8567.
- [96] Veramendi WNC, Cruvinel PE. Method for maize plants counting and crop evaluation based on multispectral images. *Comput Electron Agric* 2024;216:108470.
- [97] Cao J, Zhang Z, Luo Y, Zhang L, Zhang J, Li Z, et al. Wheat yield predictions at a county and field scale with deep learning, machine learning, and google earth engine. *Eur J Agron* 2021;123:126204.
- [98] El-Kenawy ESM, Alhussan AA, Khodadadi N, Mirjalili S, Eid MM. Predicting potato crop yield with machine learning and deep learning. *Potato Res* 2025;68:759–92.
- [99] Trentin C, Ampatzidis Y, Lacerda C, Shiratsuchi L. Tree crop yield estimation and prediction using remote sensing. *Smart Agric Technol* 2024;9:100556.
- [100] Shaikh TA, Rasool T, Venington K, Yaseen SM. The role of large language models in agriculture: harvesting the future with LLM intelligence. *Prog Artif Intell* 2025;14(2):117–64.
- [101] Shaikh TA, Rasool T, Mir WA. Fields of the future: digital transformation in smart agriculture with generative AI. *Comput Stand Interfaces* 2025;94:104005.
- [102] Tzachor A, Devare M, Richards C, Pypers P, Ghosh A, Koo J, et al. Large language models and agricultural extension services. *Nat Food* 2023;4(11):941–8.
- [103] Fominykh M, Prasolova-Førland E. Supporting career guidance and vocational training in fishery and maritime professions. *Immersive Learn Res* 2024:69–71.
- [104] Yakup MR, Doris M, Sarah NA. The application of an augmented reality offline system for rural tourism development. *Kaji Malays* 2024;42:187–214.

- [105] Wu Y, Xu M, Zhai G, Zhang W. Physical layer signal processing for XR communications and systems. *Sci China Inf Sci* 2024;67(12):221301.
- [106] Cardoso LF, Kimura BYL, Zorzal ER. Towards augmented and mixed reality on future mobile networks. *Multimedia Tools Appl* 2024;83(3):9067–102.
- [107] Maksymuk T, Gazda J, Shubyn B, Karpin O, Kapshii O, Urikova O, et al. Metaverse of things in 6G era. In: *Emerging networks digital transformation age*. Cham: Springer Nature Switzerland; 2023. p. 546–64.
- [108] Trinh B, Muntean GM. A deep reinforcement learning-based offloading scheme for multi-access edge computing-supported extended reality systems. *IEEE Trans Vehicular Technol* 2023;72(1):1254–64.
- [109] Slimani H, Jilbab A, El Mhamdi J. Interactive IoT-driven virtual reality framework for advancing digital twin applications. In: *Proceedings of the 2025 International Conference on Innovative Research Applied Science Engineering Technology*; 2025 May 15–18; Rabat, Morocco. New York City: IEEE; 2025. p. 1–7.
- [110] Kuo CY, Huang CC, Tsai CH, Shi YS, Smith S. Development of an immersive SLAM-based VR system for teleoperation of a mobile manipulator. *Comput Intell* 2021;132:103502.
- [111] Bernetti I, Borghini T, Capocchi I. Integrating virtual reality and artificial intelligence in agricultural planning. In: *Extended realities*. Cham: Springer Nature Switzerland; 2024. p. 342–50.
- [112] Singh RK, Berkvens R, Weyn M. AgriFusion: an architecture for IoT and precision agriculture survey. *IEEE Access* 2021;9:136253–83.
- [113] Fuentes D, Correia L, Costa N, Reis A, Barroso J, Pereira A, et al. SAR.IoT: secured augmented reality for IoT devices management. *Sensors* 2021;21(18):6001.
- [114] Lin X, Wang J, Wang J, Wei H, Chen M, Luo L. Picking point localization method based on semantic reasoning for complex picking scenarios. *Artif Intell Agric* 2025;15(4):744–56.
- [115] Song L, Zhao J, Li Y, Liu L, Duan J, He L, et al. Early detection of wheat powdery mildew: a multi-source in situ remote sensing approach enabled by stacked ensemble learning. *Artif Intell Agric* 2026;16(1):124–38.
- [116] Drutrer E, Miličević M, Skelin A, Bonković M. Augmented reality based sensor data visualization. In: *Proceedings of the 2022 International Conference on Software, Telecommunication and Computer Networks*; 2022 Sep 22–24; Split, Croatia. New York City: IEEE; 2022. p. 1–7.
- [117] Xing H, Deng F, Tang Y, Li Q, Zhang J. The application of augmented reality technology in urban greening plant growth state detection. *IEEE Access* 2023;11:59286–96.
- [118] Liu T, Wang J, Chen J, Zhang W, Wang Y, Zhao Y, et al. Detection of the number of wheat stems using multi-view images from smart glasses. *Comput Electron Agric* 2025;235:110370.
- [119] van Daalen T, Peller J, Balendonck J. Determining fresh tomato weight using depth images from an AR headset. *IFAC Pap* 2022;55(32):119–23.
- [120] Chai JJ, Xu JL, O'Sullivan C. Real-time detection of strawberry ripening using augmented reality and deep learning. *Sensors* 2023;23(17):7639.
- [121] Buayai P, Yok-In K, Inoue D, Nishizaki H, Makino K, Mao X. Supporting table grape berry thinning with deep neural network and augmented reality technologies. *Comput Electron Agric* 2023;213:108194.
- [122] Tamura S, Buayai P, Mao X. BerryScope: AR strawberry picking aid. In: *Proceedings of the 2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct*; 2023 Oct 16–20; Sydney, Australia. New York City: IEEE; 2023. p. 118–21.
- [123] Udekwe D, Seyyedhasani H. Virtual reality-enabled remote human-robot interaction for strawberry cultivation in greenhouses. *Comput Electron Agric* 2025;237:110567.
- [124] Goka R, Ueda K, Yamaguchi S, Kimura N, Iseya K, Kobayashi K, et al. Development of tomato harvest support system using mixed reality head mounted display. In: *Proceedings of the 2022 IEEE Global Conference on Life Sciences and Technologies*; 2022 Mar 7–9; Nara, Japan. New York City: IEEE; 2022. p. 167–9.
- [125] Sakha MM, Daiber F, Enders M, Tieben C, Kisliuk B, Kruger A, et al. Enhancing plant variety discovery process with visual trait assessment in VR. In: *Proceedings of the 2025 IEEE Conference on Virtual Reality and 3D User Interfaces*; 2025 Mar 22–26; Bremen, Germany. New York City: IEEE; 2025. p. 125–34.
- [126] Geerthik S, Senthil GA, Mohamed R, Nishaanth S. VR crop growth monitoring with AI simulations. In: *Proceedings of the 2025 International Conference on Artificial Intelligence and Machine Learning Applications for Healthcare and Internet of Things*; 2025 Mar 27–28; Chennai, India. New York City: IEEE; 2025. p. 1–6.
- [127] Nair M, Fernandez RE. Advancing mixed reality digital twins through 3D reconstruction of fresh produce. *IEEE Access* 2024;12:4315–27.
- [128] Phupattanasilp P, Tong SR. Augmented reality in the integrative internet of things: application for precision farming. *Sustainability* 2019;11(9):2658.
- [129] Ponnusamy V, Natarajan S, Ramasamy N, Clement C, Rajalingam P, Mitsunori M, et al. An IoT-enabled augmented reality framework for plant disease detection. *Rev Intell Artif* 2021;35(3):185–92.
- [130] Roopa D, Bose. Leaf's diseases and its characteristics visualization using augmented reality. In: *Proceedings of the 2022 International Conference on Computer Science and Technology*; 2022 Dec 22–24; Tirupati, India. New York City: IEEE; 2022. p. 1019–24.
- [131] Sheng H, Yao Q, Luo J, Liu Y, Chen X, Ye Z, et al. Automatic detection and counting of planthoppers on white flat plate images captured by AR glasses. *Comput Electron Agric* 2024;218:108639.
- [132] Ye Z, Liu Y, Ye F, Li H, Luo J, Guo J, et al. Automatic diagnosis of agromyzid leafminer damage levels using leaf images captured by AR glasses. *J Integr Agric* 2025;24(9):3559–73.
- [133] Pawar S, Dupare K, Ganvir T, Chore D. IoT-based home automation: augmented reality enabled plant watering system. In: *2025 International Conference on Interdisciplinary Approaches Technology Management Society Innovation*; 2025; Gwalior, India. New York City: IEEE; 2025. p. 1–6.
- [134] Slimani H, El Mhamdi J, Jilbab A. Integrating augmented reality and IoT sensors for advanced real-time greenhouse control. In: *Farhaoui Y, Herawan T, Imoize AL, Allaoui AE, editors. Intersection of artificial intelligence and data science: cutting-edge concepts and smart environment applications*. Cham: Springer Nature Switzerland; 2025. p. 113–9.
- [135] Romano E, Bisaglia C, Lazzari A, Filisetti A, Premoli E, Brambilla M. Augmented reality for the management of microclimate variability in greenhouses. In: *Proceedings of the 2023 IEEE International Workshop on Metrology for Agriculture and Forestry*; 2023 Nov 6–8; Pisa, Italy. New York City: IEEE; 2023. p. 811–4.
- [136] Slimani H, El Mhamdi J, Jilbab A. Real-time greenhouse management using IoT, digital twin, and augmented reality for optimal control and decision-making. *Acta IMEKO* 2025;14(2):1–15.
- [137] Slob N, Hurst W, van de Zedde R, Tekinerdogan B. Virtual reality-based digital twins for greenhouses: a focus on human interaction. *Comput Electron Agric* 2023;208:107815.
- [138] Do TH, Pham VMT, Hoang HH, Bui TD, Yamada K. Combining LoRaWAN with AR in IoT-enabled smart farm monitoring. In: *Lecture notes in networks and systems*; vol. 882. Cham: Springer Science and Business Media Deutschland; 2024. p. 38–49.
- [139] Maasthi MJ. An interactive augmented reality application for agriculture using Unity. In: *Proceedings of the International Conference on Communication Systems and Networks*; 2025 Mar 20–22; Bengaluru, India. New York City: IEEE; 2025. p. 1371–3.
- [140] Zheng M, Campbell AG. Location-based augmented reality in-situ visualization applied for agricultural fieldwork navigation. In: *Proceedings of the 2019 IEEE International Symposium on Mixed and Augmented Reality Adjunct*; 2019 Oct 14–18; Beijing, China. New York City: IEEE; 2019. p. 93–7.
- [141] Lopes WAC, Fernandes JCL, Antunes SN, Fernandes ME, Nääs IA, Vendrametto O, et al. Augmented reality applied to identify aromatic herbs using mobile devices. *AgriEngineering* 2024;6(3):2824–44.
- [142] Katsaros A, Keramopoulos E. FarmAR: a farmer's augmented reality application based on semantic web. In: *Proceedings of the 2017 South East European Design Automation and Computer Engineering Conference*; 2017 Apr 24–26; Athens, Greece. New York City: IEEE; 2017. p. 1–6.
- [143] Carruth DW, Hudson C, Fox AAA, Deb S. User interface for an immersive virtual reality greenhouse for training precision agriculture. In: *Chen, JYC, Fragomeni G, editors. Virtual, augmented and mixed reality. Industrial and everyday life applications. HCII 2020. Lecture notes in computer science*; vol. 12191. Cham: Springer; 2020. p. 35–46.
- [144] Kim R, Kim J, Lee I, Yeo U, Lee S, Decano-Valentin C. Development of three-dimensional visualisation technology of the aerodynamic environment in a greenhouse, part 2: development of an educational VR simulator. *Biosyst Eng* 2021;207:12–32.
- [145] Huuskonen J, Oksanen T. Soil sampling with drones and augmented reality in precision agriculture. *Comput Electron Agric* 2018;154:25–35.
- [146] Nwobodo OJ, Wereszczyński K, Kuaban GS, Skurowski P, Cyran KA. An adaptation of Fitts' law for performance evaluation and optimization of augmented reality interfaces. *IEEE Access* 2024;12:169614–27.
- [147] Agalave HR. Effect of environmental factors on productivity of crop. *Int J Bot Stud* 2017;2:14–6.
- [148] Li B, Wang Y, Rong L, Zheng W. Research progress on animal environment and welfare. *Anim Res One Health* 2023;1:78–91.
- [149] Zhao Z, Yang W, Chinthammit W, Rawnsley R, Neumeyer P, Cahoon S. A new approach to utilize augmented reality on precision livestock farming. In: *Proceedings of the 27th International Conference on Artificial Reality and Telexistence and 22nd Eurographics Symposium on Virtual Environments*; 2017 Nov 22–24; Adelaide, Australia. Goslar: Eurographics Association; 2017. p. 185–8.
- [150] Pinna D, Sara G, Todde G, Atzori AS, Artizzu V, Spano LD, et al. Advancements in combining electronic animal identification and augmented reality technologies in digital livestock farming. *Sci Rep* 2023;13:18282.
- [151] Sara G, Todde G, Caria M. Assessment of video see-through smart glasses for augmented reality to support technicians during milking machine maintenance. *Sci Rep* 2022;12:15729.
- [152] Arkan I, Ayav T, Seçkin AÇ, Soygazi F. Estrus detection and dairy cow identification with cascade deep learning for augmented reality-ready livestock farming. *Sensors* 2023;23(24):9795.
- [153] Caria M, Sara G, Todde G, Polese M, Pazzona A. Exploring smart glasses for augmented reality: a valuable and integrative tool in precision livestock farming. *Animals* 2019;9(11):903.
- [154] Caria M, Todde G, Sara G, Piras M, Pazzona A. Performance and usability of smartglasses for augmented reality in precision livestock farming operations. *Appl Sci* 2020;10(7):2318.
- [155] Baltzer D, Douglas S, Haunert JH, Dehbi Y, Tiemann I. Smart glasses in the chicken barn: enhancing animal welfare through mixed reality. *Smart Agric Technol* 2025;10:100786.

- [156] Kim R, Kim J, Lee I, Yeo U, Lee S. Development of a VR simulator for educating CFD-computed internal environment of piglet house. *Biosyst Eng* 2019;188:243–64.
- [157] Xi M, Adcock M, McCulloch J. An end-to-end augmented reality solution to support aquaculture farmers with data collection, storage, and analysis. In: *Proceedings of the 17th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and Its Applications in Industry*; 2019 Nov 14–16; Brisbane, Australia. New York City: ACM; 2019. p. 1–2.
- [158] Rahman A, Xi M, Dabrowski JJ, McCulloch J, Arnold S, Rana M, et al. An integrated framework of sensing, machine learning, and augmented reality for aquaculture prawn farm management. *Aquacult Eng* 2021;95:102192.
- [159] Napier T, Lee I. Using mobile-based augmented reality and object detection for real-time abalone growth monitoring. *Comput Electron Agric* 2023;207:107744.
- [160] Wang Z, Xun Y, Wang Y, Yang Q. Review of smart robots for fruit and vegetable picking. *Int J Agric Biol Eng* 2022;15:33–54.
- [161] Cho W, Nguyen NV, Hayashi K. Experimental evaluation of remote semi-autonomous tractors in paddy fields. *Appl Eng Agric* 2025;41:179–93.
- [162] Stotko P, Krumpfen S, Schwarz M, Lenz C, Behnke S, Klein R, et al. A VR system for immersive teleoperation and live exploration with a mobile robot. In: *Proceedings of the 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems*; 2019 Oct 14–18; Macau, China. New York City: IEEE; 2019. p. 3630–7.
- [163] Jaiswal S, Aman R, Sopanen J, Mikkola A. Real-time multibody model-based heads-up display unit of a tractor. *IEEE Access* 2021;9:57645–57.
- [164] Kaizu Y, Choi J. Development of a tractor navigation system using augmented reality. *Eng Agric Environ Food* 2012;5(3):96–101.
- [165] Huuskonen J, Oksanen T. Augmented reality for supervising multirobot system in agricultural field operation. *IFAC Pap* 2019;52(30):367–72.
- [166] Santana-Fernández J, Gómez-Gil J, del-Pozo-San-Cirilo L. Design and implementation of a GPS guidance system for agricultural tractors using augmented reality technology. *Sensors* 2010;10(11):10435–47.
- [167] Abdulsalam M, Aouf N. VitRob pipeline: a seamless teleoperation pipeline for advanced virtual reality–robot interface applied for precision agriculture. In: *Proceedings of the 2023 IEEE International Conference on Robotics and Biomimetics*; 2023 Dec 4–9; Koh Samui, Thailand. New York City: IEEE; 2023. p. 1–6.
- [168] Lohan SK, Narang MK, Raghuvirsinh P, Kumar S, Singh LP. Development and field evaluation of a VR/AR-based remotely controlled two-wheel paddy transplanter. *J Field Robot* 2024;41(8):2732–48.
- [169] Chen Y, Zhang B, Zhou J, Wang K. Real-time 3D unstructured environment reconstruction utilizing VR and Kinect-based immersive teleoperation. *Comput Electron Agric* 2020;175:105579.
- [170] Wang D, Zhang B, Zhou J, Xiong Y, Liu L, Tan Q. Three-dimensional mapping and immersive human–robot interfacing utilizing Kinect-style depth cameras and virtual reality. *J Field Robot* 2024;41(7):2413–26.
- [171] Wang Y, Zhou J, Zheng Y, Rottok LT, Jiang Z, Sun J. VR map construction for orchard robot teleoperation based on dual-source positioning and sparse point cloud segmentation. *Comput Electron Agric* 2024;224:109187.
- [172] Faust K, Casteel C, McGehee DV, Peek-Asa C, Rohlman DS, Ramirez M. Examination of realism in a high-fidelity tractor driving simulator. *J Agric Saf Health* 2020;26(4):123–37.
- [173] Meusel C, Kieu D, Gilbert S, Luecke G, Gilmore B, Kelly N, et al. Evaluating operator harvest technology within a high-fidelity combine simulator. *Comput Electron Agric* 2018;148:309–21.
- [174] Meusel C, Grimm C, Starkey J, Gilbert SB, Luecke G, et al. The importance of operator knowledge in evaluating virtual reality cue fidelity. *Comput Electron Agric* 2019;160:179–87.
- [175] Luecke GR. GREENSPACE: virtual reality interface for combine operator training. *Presence* 2012;21(3):245–53.
- [176] Ojados Gonzalez D, Martin-Goriz B, Ibarra Berrocal I, Macian Morales A, Adolfo Salcedo G, Miguel HB. Development and assessment of a tractor driving simulator with immersive virtual reality for training to avoid occupational hazards. *Comput Electron Agric* 2017;143:111–8.
- [177] V A, S BK, V DS, R M. Revolutionizing agriculture: empowering farmers through drone training simulations. In: *Proceedings of the 2024 10th International Conference on Advanced Computing Communication Systems*; 2024 Mar 14–15; Coimbatore, India. New York City: IEEE; 2024. p. 822–6.
- [178] Shi L, Shi G, Qiu H. General review of intelligent agriculture development in China. *China Agric Econ Rev* 2018;11:39–51.
- [179] Karunathilake EMBM, Le AT, Heo S, Chung YS, Mansoor S. The path to smart farming: innovations and opportunities in precision agriculture. *Agriculture* 2023;13(8):1593.
- [180] Erickson B, Fausti S, Clay D, Clay S. Knowledge, skills, and abilities in the precision agriculture workforce: an industry survey. *Nat Sci Educ* 2018;47(1):180010.
- [181] Strousopoulos P, Troussas C, Papakostas C, Krouska A, Sgouropoulou C. Revolutionizing agricultural education with virtual reality and gamification. In: Kabassi K, Mylonas P, Caro J, editors. *Novel intelligent digital systems: proceedings of the 3rd International Conference (NiDS 2023)*; 2023 Jun 29–Jul 1; Athens, Greece. Cham: Springer Nature Switzerland; 2023. p. 67–80.
- [182] Ario MK, Hasani MF, Balqis K, Carment M. HoloFarm: enhancing agricultural learning through immersive technology. *Softw Impacts* 2025;24:100768.
- [183] Lasseur R, Laurenson S, Ali M, Loh I, Mackay M. Designing profitable and climate-smart farms using virtual reality. *PLoS One* 2023;18(6):e0286723.
- [184] Parikh T, Egendorf SP, Murray I, Jamali A, Yee B, Lin S. Greening the virtual smart city: accelerating peer-to-peer learning in urban agriculture with virtual reality environments. *Front Sustain Cities* 2022;3:815937.
- [185] Schütz A, Kurz K, Busch G. Virtual farm tours—virtual reality glasses and tablets are suitable tools to provide insights into pig husbandry. *PLoS One* 2022;17(1):e0261248.
- [186] Yang Y, Fan Y, Sun R. A human–computer interaction system for agricultural tools museum based on virtual reality technology. *Adv Multimedia* 2019;2019:2659313.
- [187] Peladarios N, Piromalis D, Cheimaras V, Tserepas E, Munteanu RA, Papageorgas P, et al. Enhancing smart agriculture by implementing digital twins: a comprehensive review. *Sensors* 2023;23(16):7128.
- [188] Sara G, Pinna D, Todde G, Caria M. Augmented reality glasses applied to livestock farming: potentials and perspectives. *AgriEngineering* 2024;6(2):1859–69.
- [189] Xie J, Chai JJK, Xu JL. Trends of augmented reality for agri-food applications. *Sensors* 2022;22(21):8333.
- [190] Klerkx L, Jakku E, Labarthe P. A review of social science on digital agriculture, smart farming and agriculture 4.0: new contributions and a future research agenda. *NJAS Wageningen, J Life Sci* 2019;90–91:1–16.
- [191] Greig JA, Colclasure B, Rampold S, Ruth T, Granberry T. Enhancing agricultural education through virtual reality: facilitation, application, reflection, and measurement in the classroom. *Adv Agric Dev* 2024;5:64–80.
- [192] Weichelt B, Heimonen T, Pilz M, Yoder A, Bendixsen C. An argument against cross-platform development: lessons from an augmented reality app prototype for rural emergency responders. *JMIR Mhealth Uhealth* 2019;7(3):e12207.
- [193] Arulmozhi E, Deb NC, Tamrakar N, Kang DY, Kang MY, Kook J, et al. From reality to virtuality: revolutionizing livestock farming through digital twins. *Agriculture* 2024;14(12):2231.
- [194] Yang X, Shu L, Chen J, Ferrag MA, Wu J, Nurellari E, et al. A survey on smart agriculture: development modes, technologies, and security and privacy challenges. *IEEE/CAA J Autom Sin* 2021;8:273–302.
- [195] Huang S, Isaac M, Gao Y, Dong Y, Zhang J, Hu B. Blockchain-empowered UAV-assisted computation resource allocation in smart farm services platform. *IEEE Access* 2025;13:92675–89.
- [196] Biener V, Winston FJ, Schmalstieg D, Plopski A. Long-term experiences from working with extended reality in the wild. In: *Proceedings of the 2025 IEEE International Symposium on Mixed and Augmented Reality*; 2025 Oct 8–12; Daejeon, Korea. New York City: IEEE; 2025. p. 667–76.
- [197] Halman J, Tencer S, Siemiński M. Artificial intelligence and extended reality in the training of vascular surgeons: a narrative review. *Med Sci* 2025;13(3):126.
- [198] Balaska V, Adamidou Z, Vryzas Z, Gasteratos A. Sustainable crop protection via robotics and artificial intelligence solutions. *Machines* 2023;11(8):774.
- [199] Whig A, Whig P, Gupta SK. Harnessing XR technologies and deep learning for environmental sustainability. In: Gupta SK, Maurya N, Le DN, Mzili T, editors. *Exploring the impact of extended reality technologies to promote environmental sustainability*. Cham: Springer Nature Switzerland; 2025. p. 1–20.
- [200] Vullaganti N, Ram BG, Sun X. Precision agriculture technologies for soil site-specific nutrient management: a comprehensive review. *Artif Intell Agric* 2025;15:147–61.
- [201] Gami SJ. Data science for environmental impact assessment using XR. In: Gupta SK, Maurya N, Le DN, Mzili T, editors. *Exploring the impact of extended reality technologies to promote environmental sustainability*. Cham: Springer Nature Switzerland; 2025. p. 181–98.
- [202] Lo TTS, Chen Y, Lai TY, Goodman A. Phygital workspace: a systematic review in developing a new typological work environment using XR technology to reduce carbon footprint. *Front Built Environ* 2024;10:1370423.
- [203] Richterova E, Richter M. Agricultural eco-efficiency vs efficiency in the EU-27. *Equilib Q J Econ Econ Policy* 2025;20(4):1425–58.
- [204] Lazaroiu G, Andronie M, Uță C, Hurlui I. Trust management in organic agriculture: sustainable consumption behavior, environmentally conscious purchase intention, and healthy food choices. *Front Public Health* 2019;7:340.
- [205] Jareño F, Tolentino M, del Valle FM. Energy-agriculture market linkages: asymmetric effects in the context of the Russia–Ukraine conflict. *Oeconomia Copernic* 2025;16(1):163–96.